

(12) **United States Patent**
Ginn

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(54) **SYSTEMS FOR SACROILIAC JOINT
STABILIZATION**

A61B 17/8858 (2013.01); *A61B 34/20*
(2016.02); *A61F 2/4455* (2013.01); *A61F*
2/4601 (2013.01); *A61F 2/4611* (2013.01);
A61B 2576/00 (2013.01);

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(US)

(Continued)

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(58) **Field of Classification Search**

CPC *A61F 2002/30995*; *A61F 2/4455*; *A61B*
17/1671; *A61B 17/1757*
See application file for complete search history.

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(US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 1044 days.

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(21) Appl. No.: **17/463,831**

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(Continued)

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Primary Examiner — Lynnsy M Summitt

(74) Attorney, Agent, or Firm — Francis Law Group

Related U.S. Application Data

(63) Continuation-in-part of application No. 13/857,977,
filed on Apr. 5, 2013, now Pat. No. 11,273,042,
(Continued)

(57)

ABSTRACT

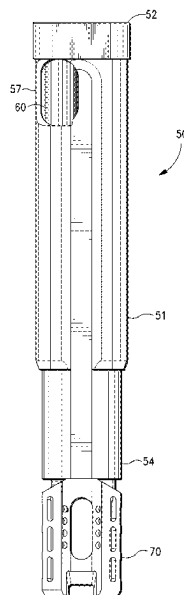
Systems are described for stabilizing a dysfunctional sacroiliac (SI) joint of a subject. The systems include a tool assembly and a defect creation assembly, and a prosthesis. The tool assembly is adapted to create a pilot SI joint opening in the dysfunctional SI joint; portions of which being disposed in the sacrum and ilium bone structures. The prosthesis is sized and configured to be press-fit into the pilot SI joint opening, wherein the pilot SI joint opening transitions to a larger post-prosthesis insertion SI joint opening and the prosthesis is securely engaged to the sacrum and ilium bone structures. The system optionally includes an image capture apparatus adapted to capture images reflecting positions and/or orientations of the tool assembly when disposed in the subject's body.

(51) **Int. Cl.**
A61F 2/44 (2006.01)
A61B 5/055 (2006.01)

(Continued)

(52) **U.S. Cl.**
CPC *A61F 2/30988* (2013.01); *A61B 5/055*
(2013.01); *A61B 5/062* (2013.01); *A61B 6/032*
(2013.01); *A61B 6/485* (2013.01); *A61B*
8/0841 (2013.01); *A61B 17/1604* (2013.01);
A61B 17/1664 (2013.01); *A61B 17/1671*
(2013.01); *A61B 17/70* (2013.01); *A61B*
17/7055 (2013.01); *A61B 17/7074* (2013.01);

4 Claims, 33 Drawing Sheets



Related U.S. Application Data

which is a continuation of application No. 13/192,289, filed on Jul. 27, 2011, now abandoned.

(60) Provisional application No. 61/368,233, filed on Jul. 27, 2010.

(51) **Int. Cl.**

A61B 5/06 (2006.01)
A61B 6/00 (2006.01)
A61B 6/03 (2006.01)
A61B 8/08 (2006.01)
A61B 17/16 (2006.01)
A61B 17/70 (2006.01)
A61B 17/88 (2006.01)
A61B 34/20 (2016.01)

A61F 2/30 (2006.01)

A61F 2/46 (2006.01)

(52) **U.S. Cl.**

CPC *A61F 2002/30121* (2013.01); *A61F 2002/30123* (2013.01); *A61F 2002/30166* (2013.01); *A61F 2002/30593* (2013.01); *A61F 2002/30995* (2013.01)

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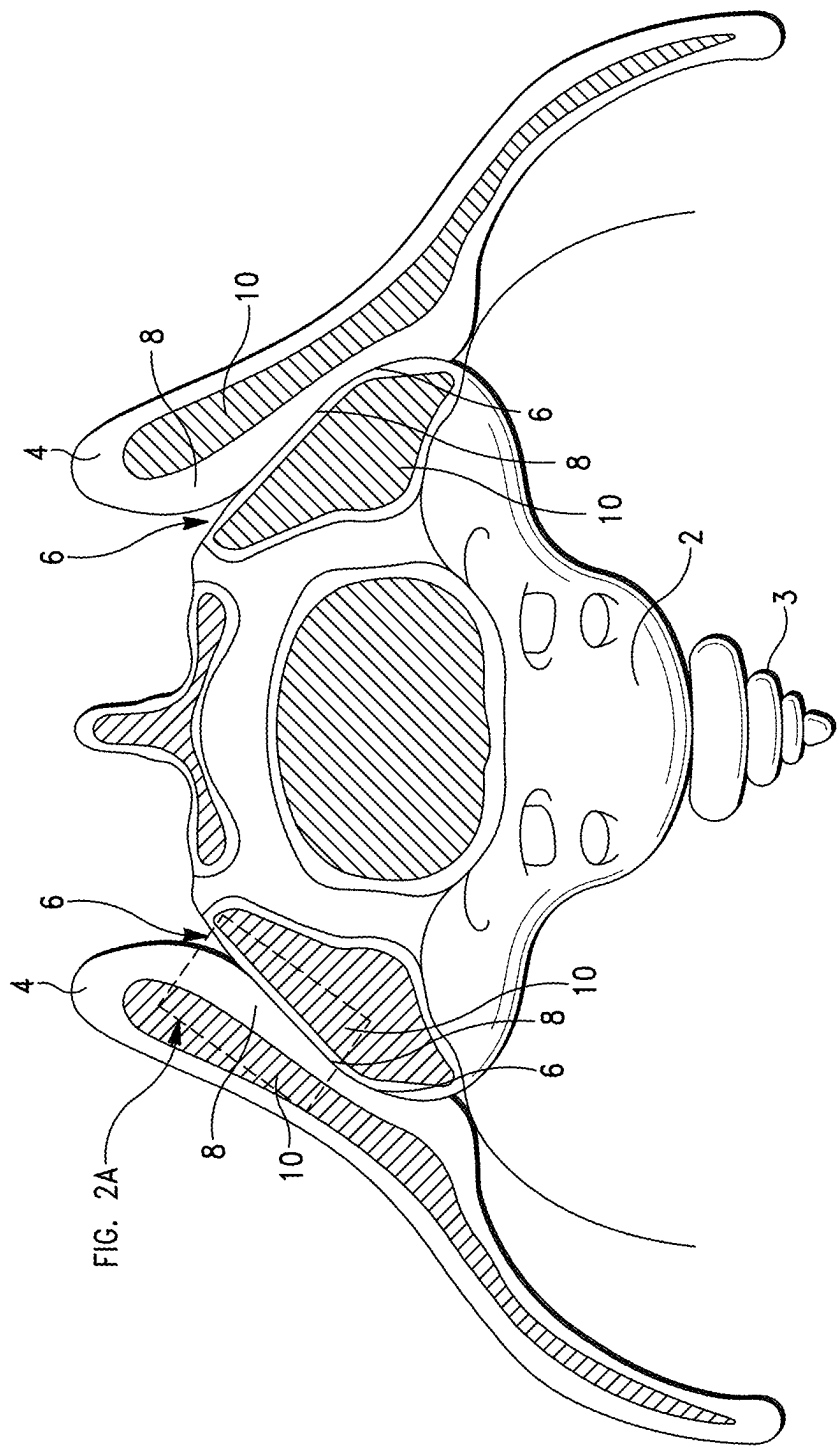


FIG. 1A

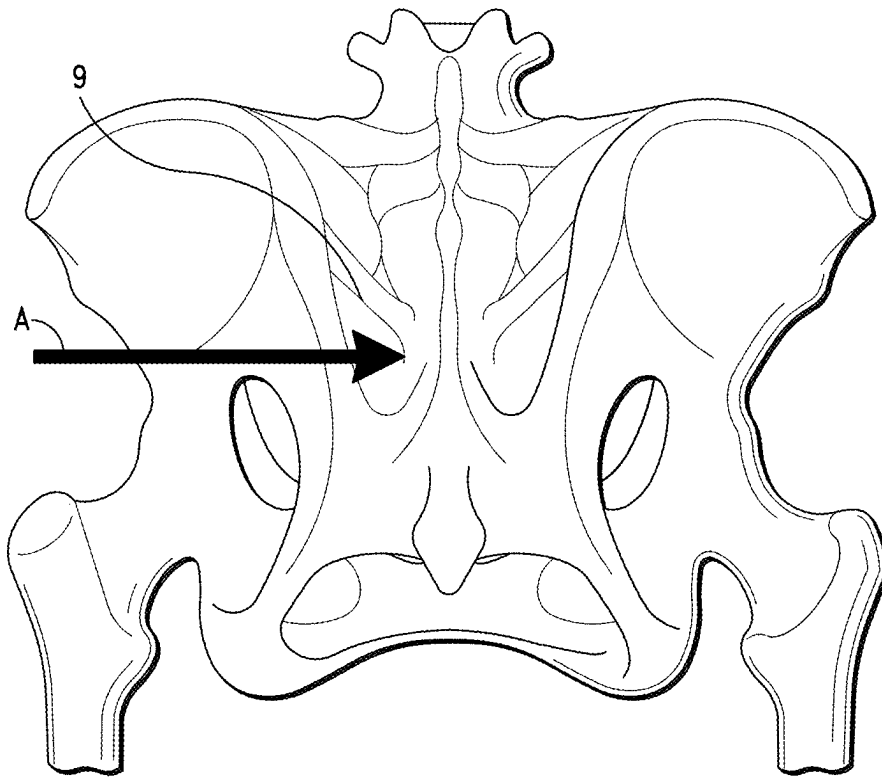


FIG. 1B

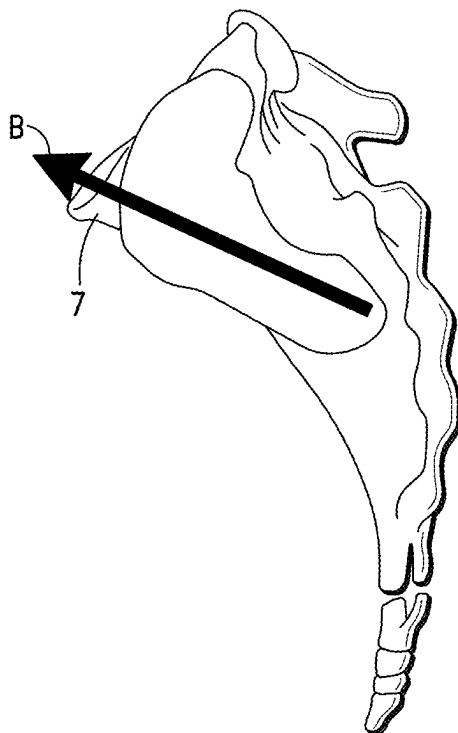


FIG. 1C

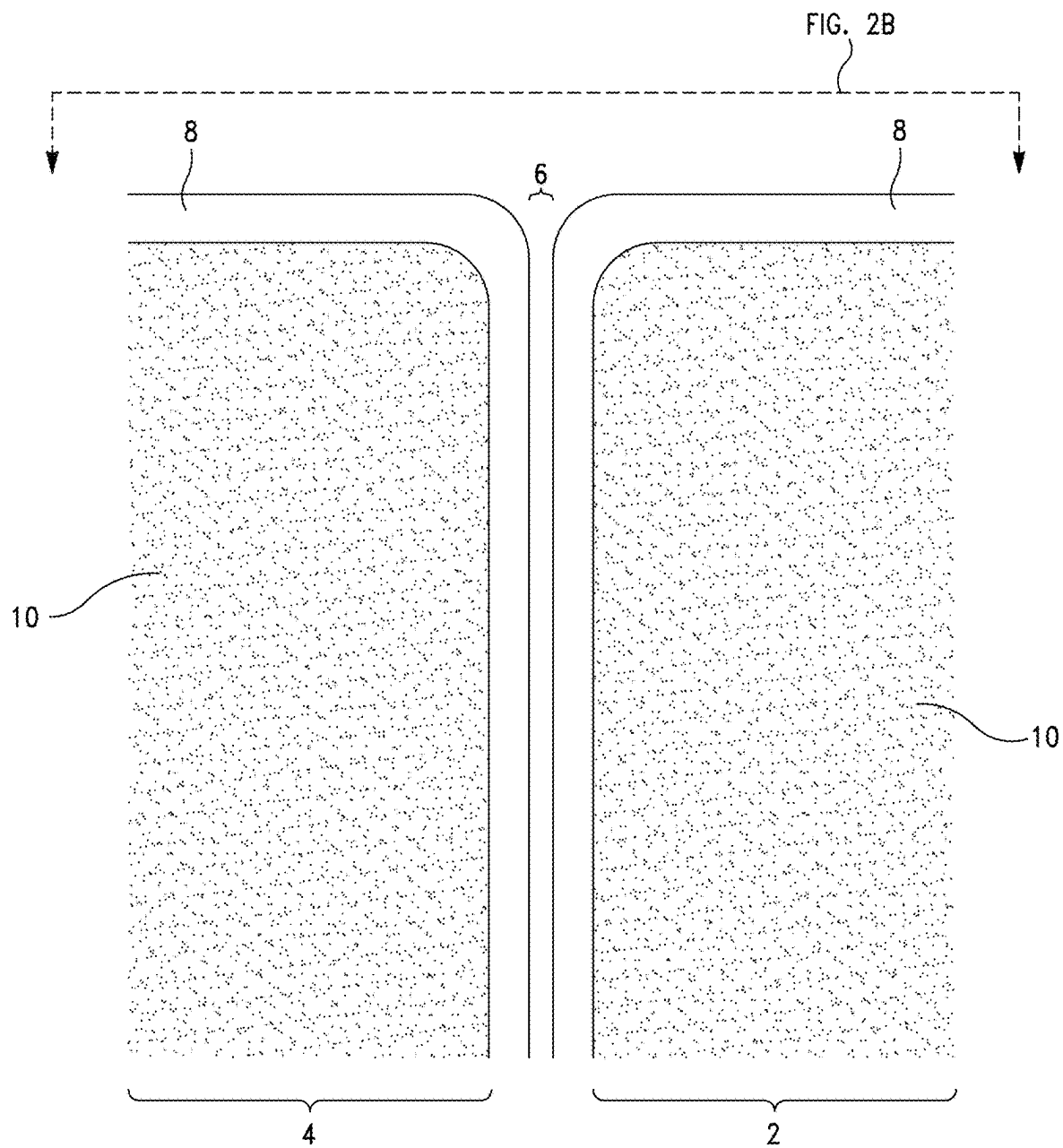
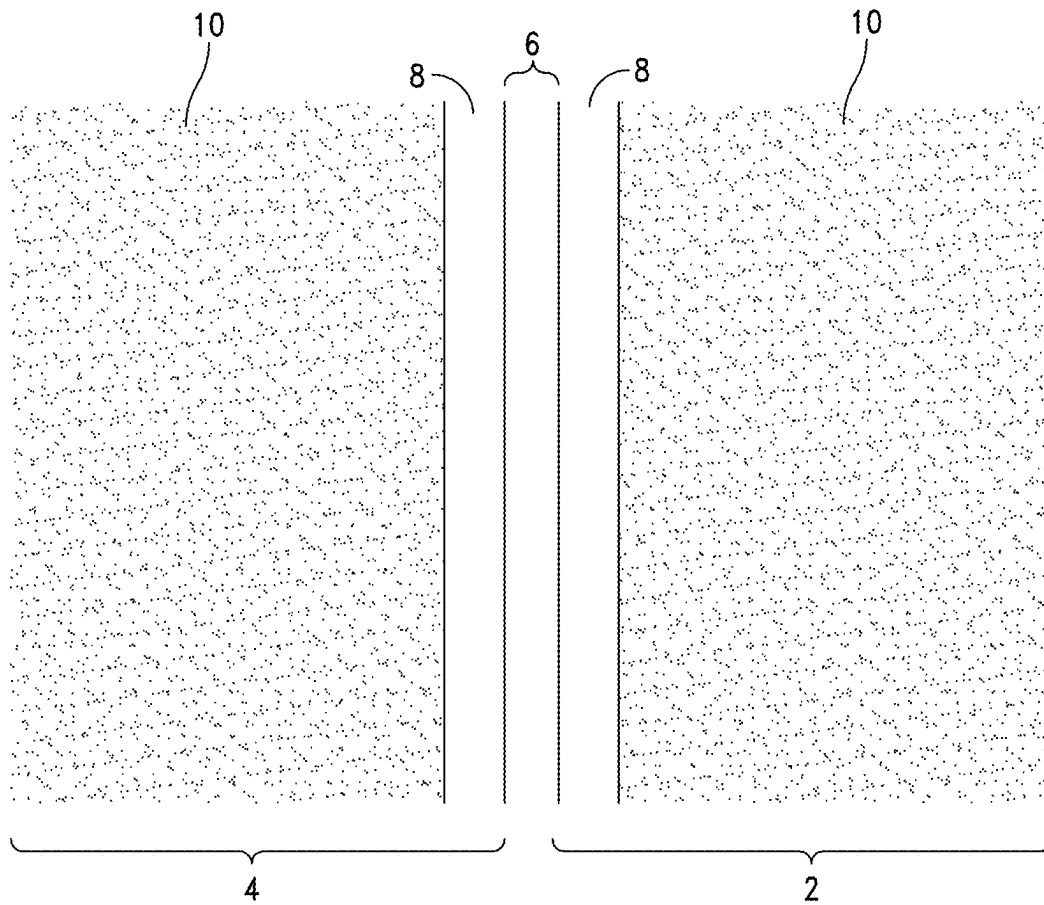


FIG. 2A

*FIG. 2B*

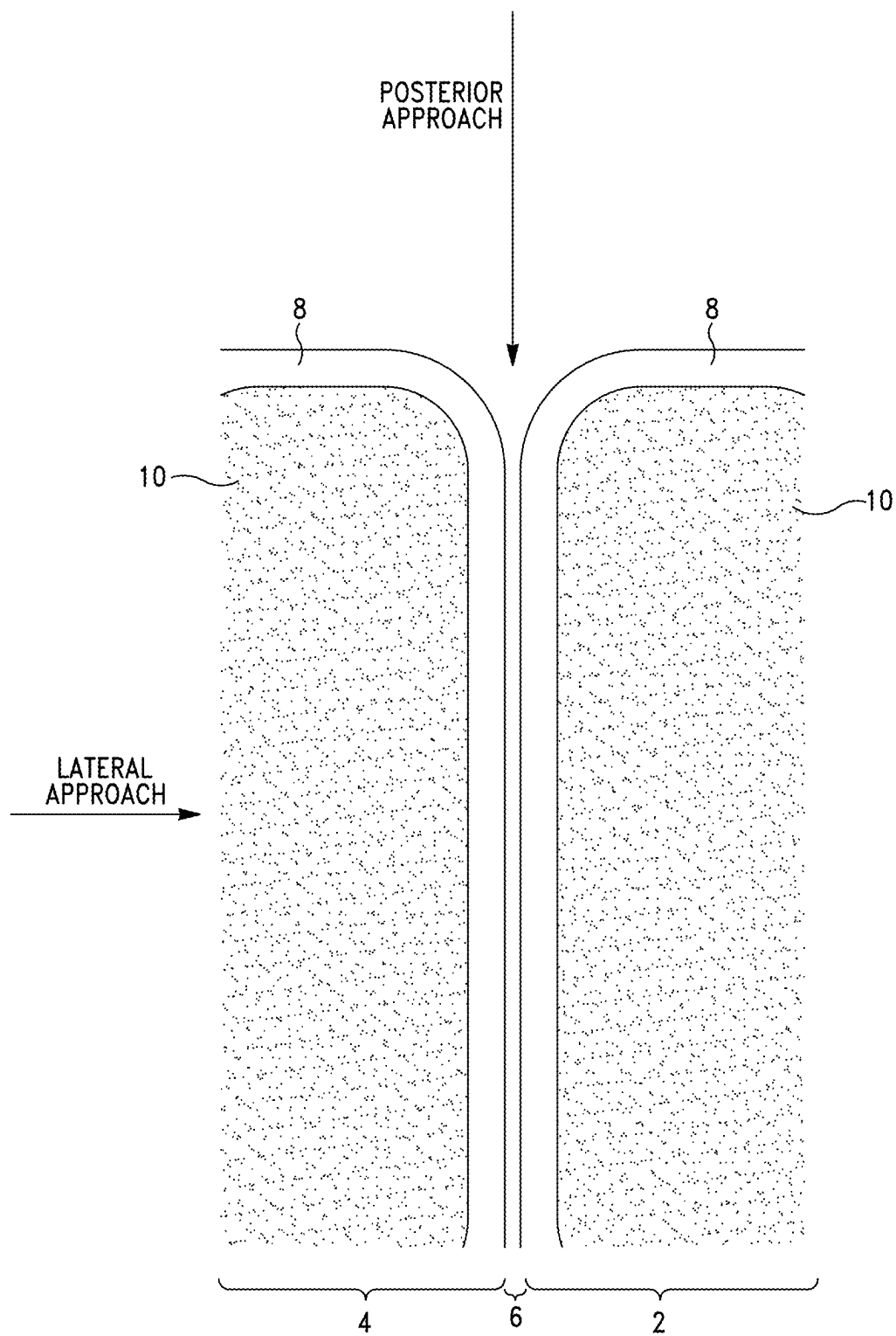
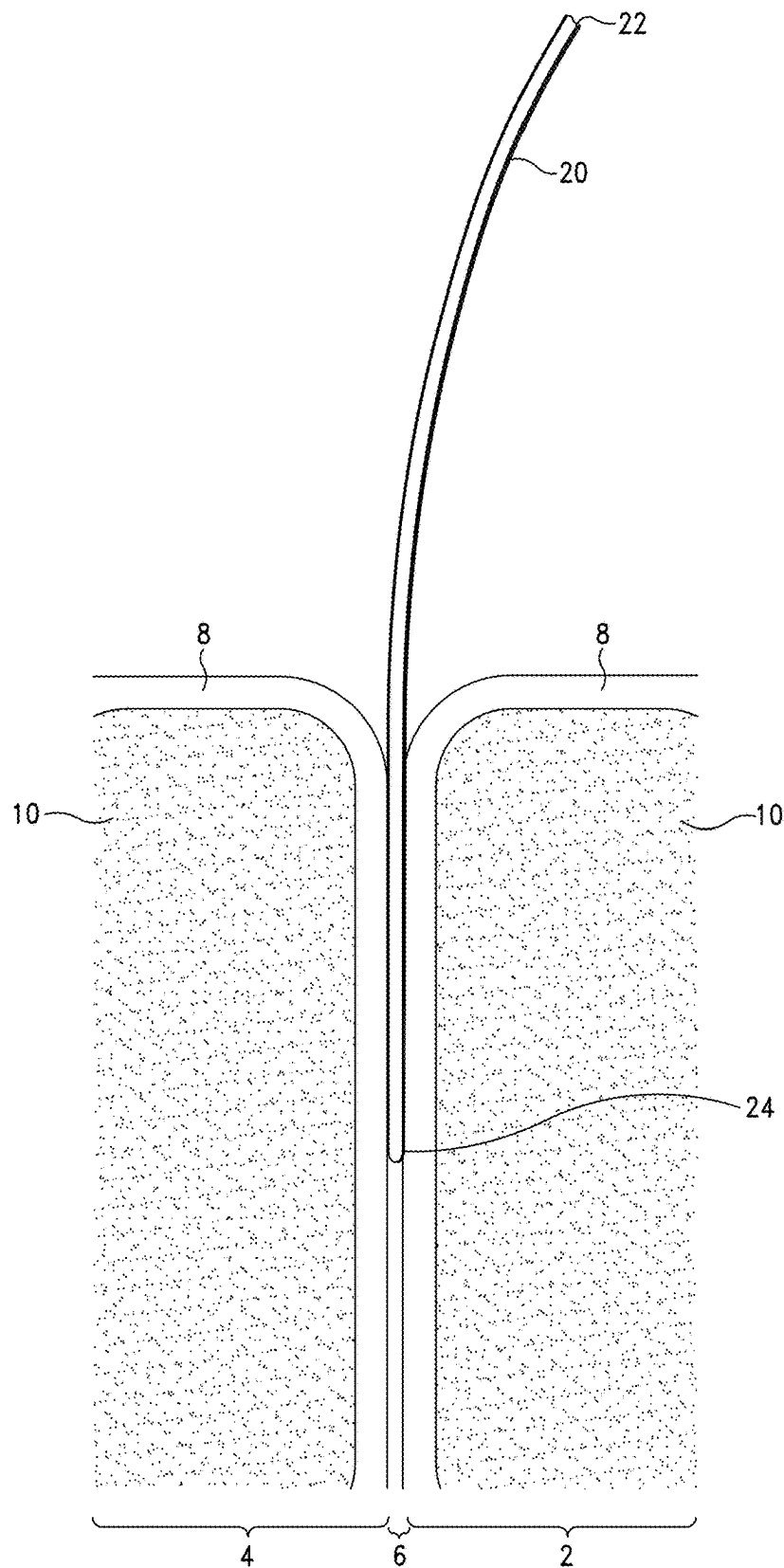


FIG. 2C

*FIG. 3A*

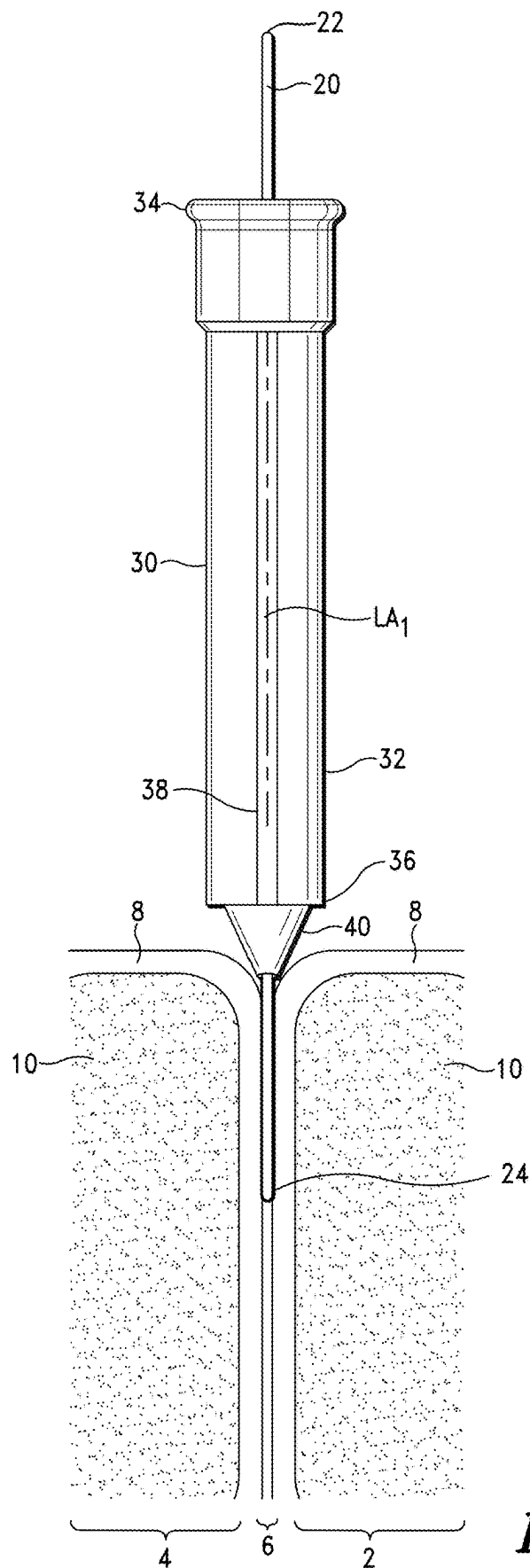


FIG. 3B

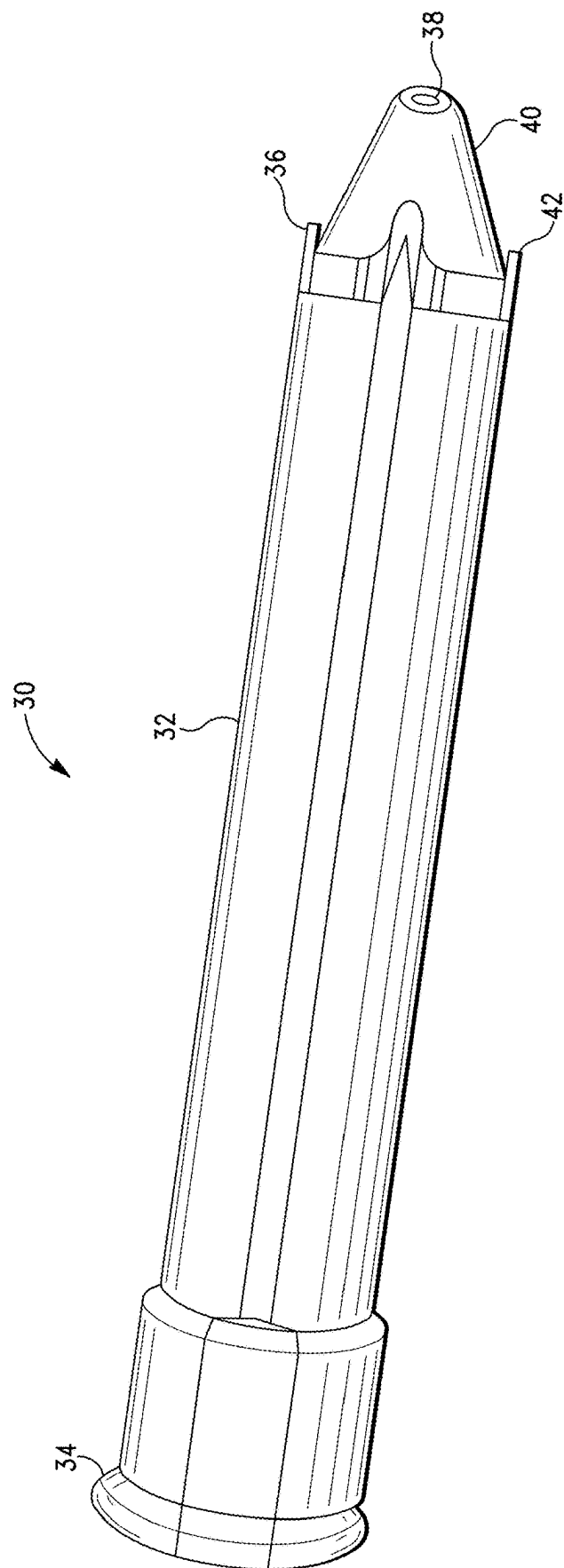


FIG. 3C

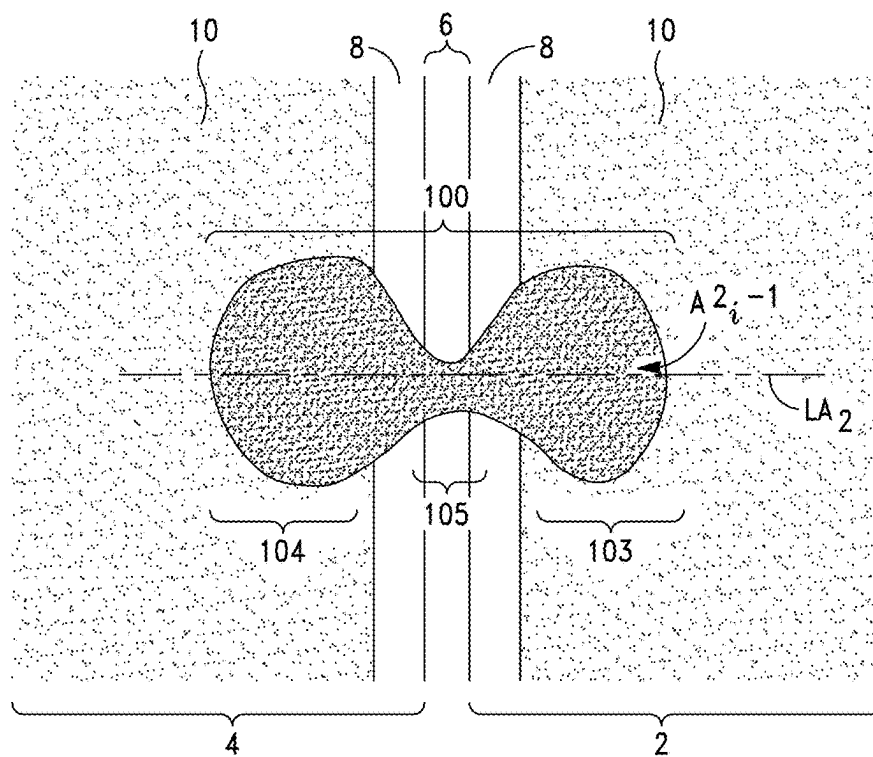


FIG. 4A

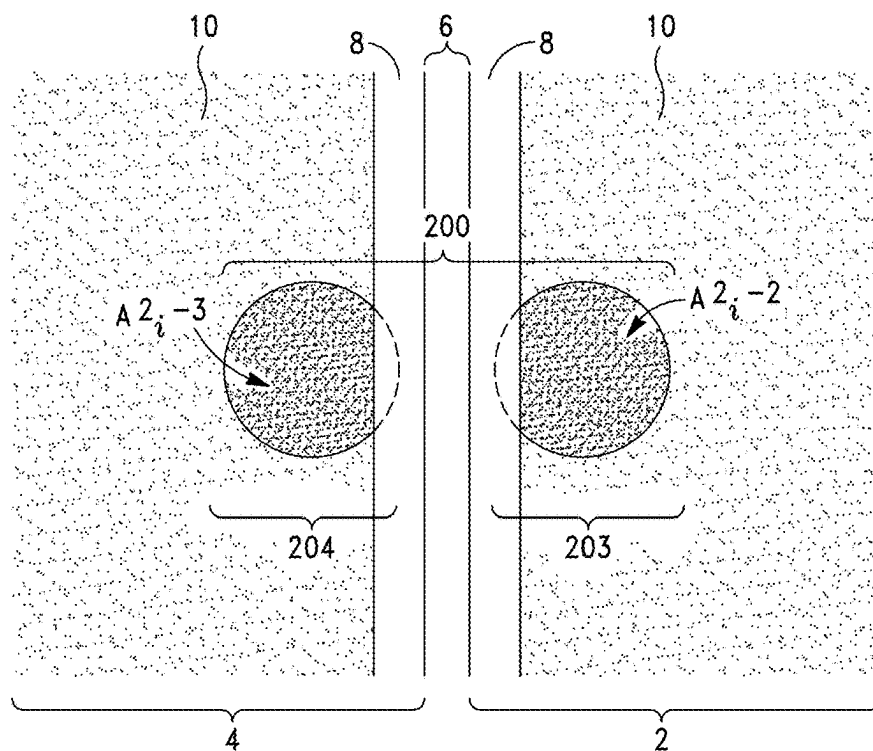


FIG. 4B

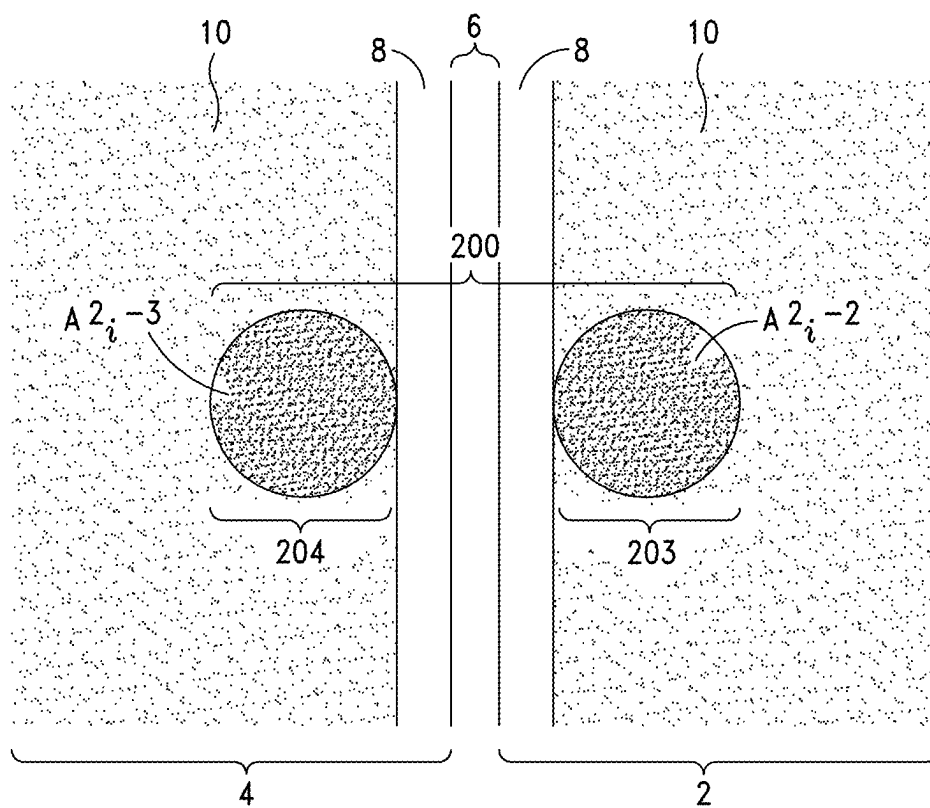


FIG. 4C

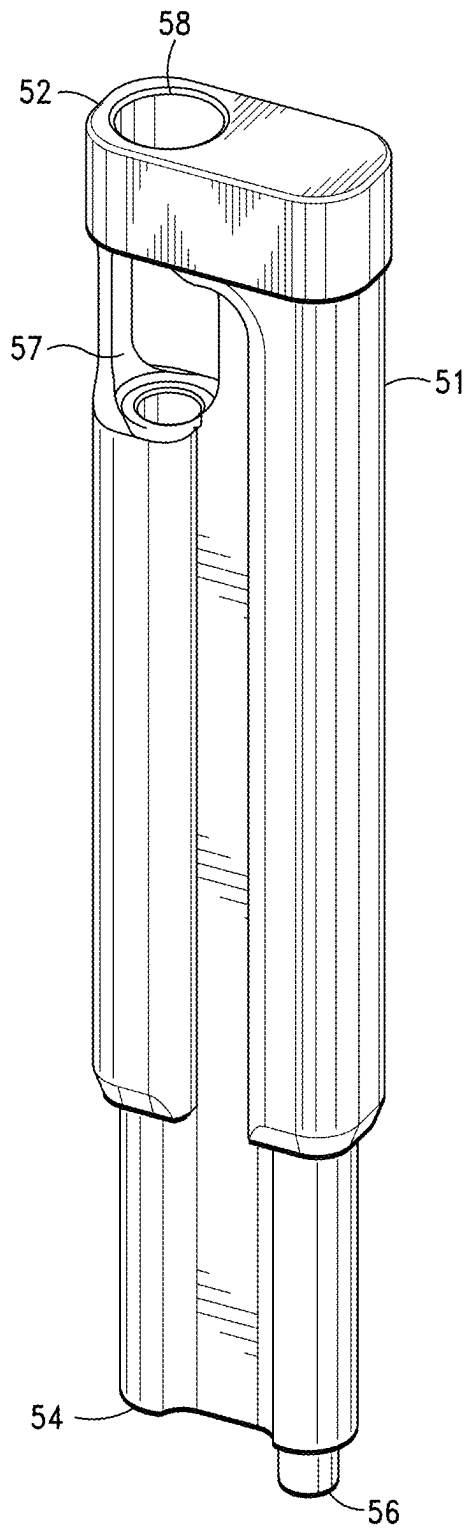


FIG. 5A

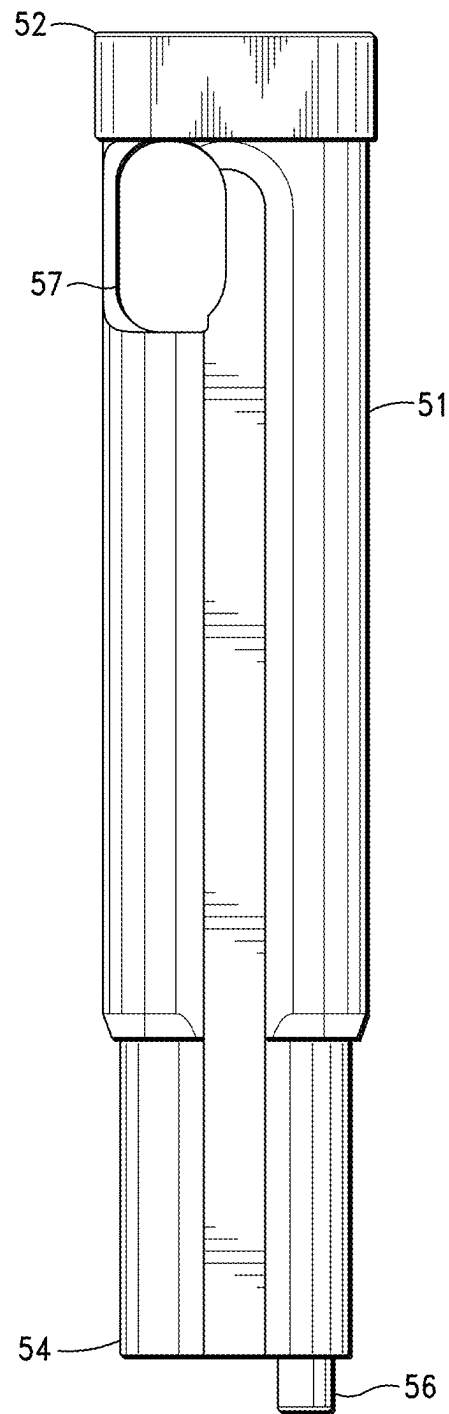


FIG. 5B

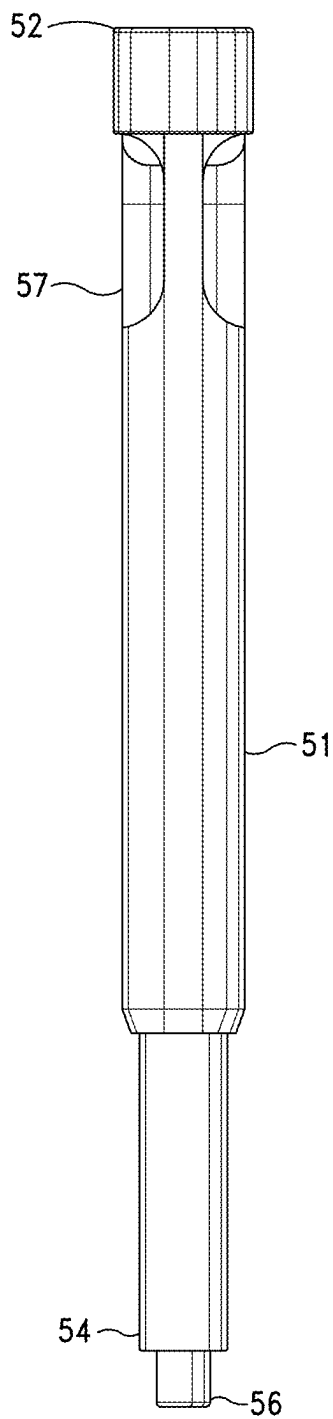


FIG. 5C

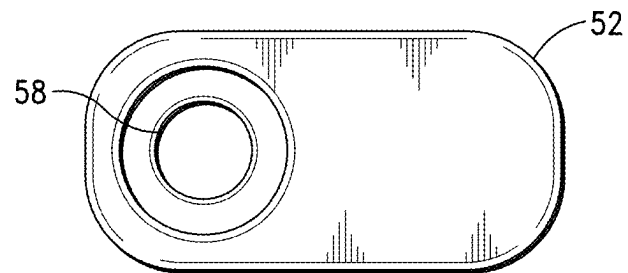


FIG. 5D

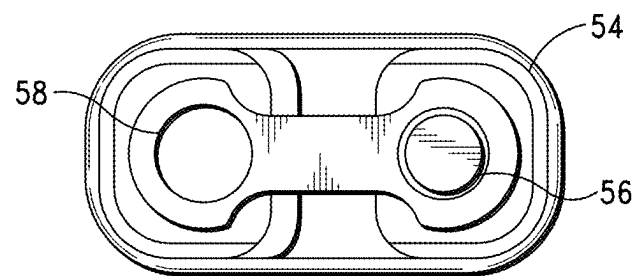


FIG. 5E

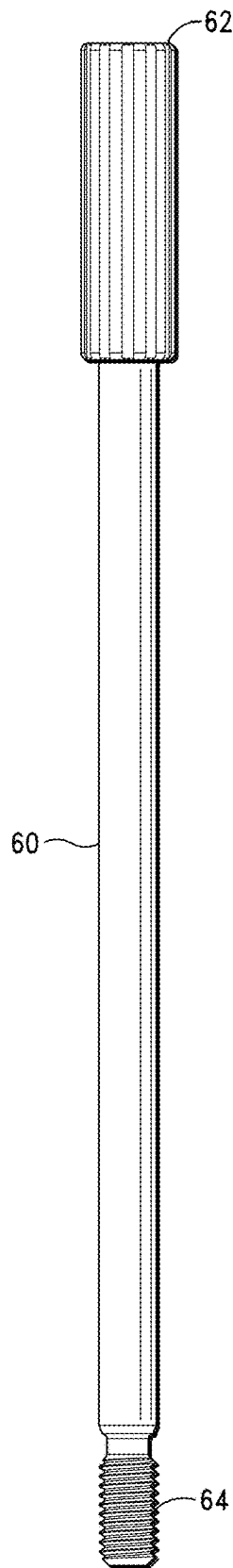


FIG. 5F

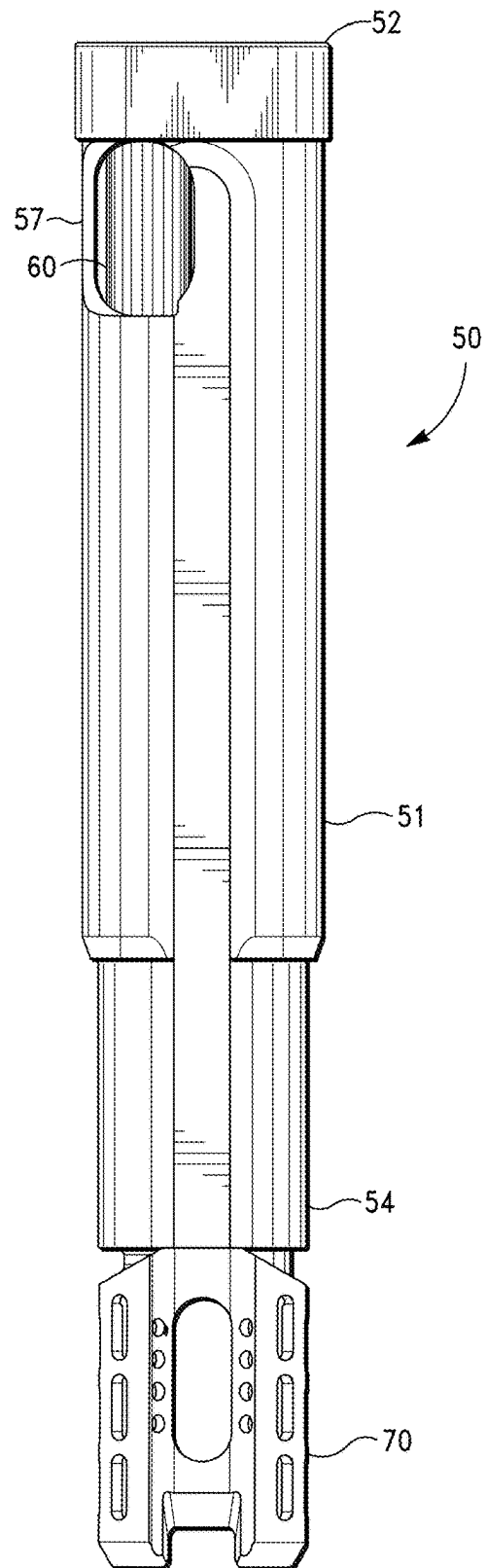


FIG. 5G

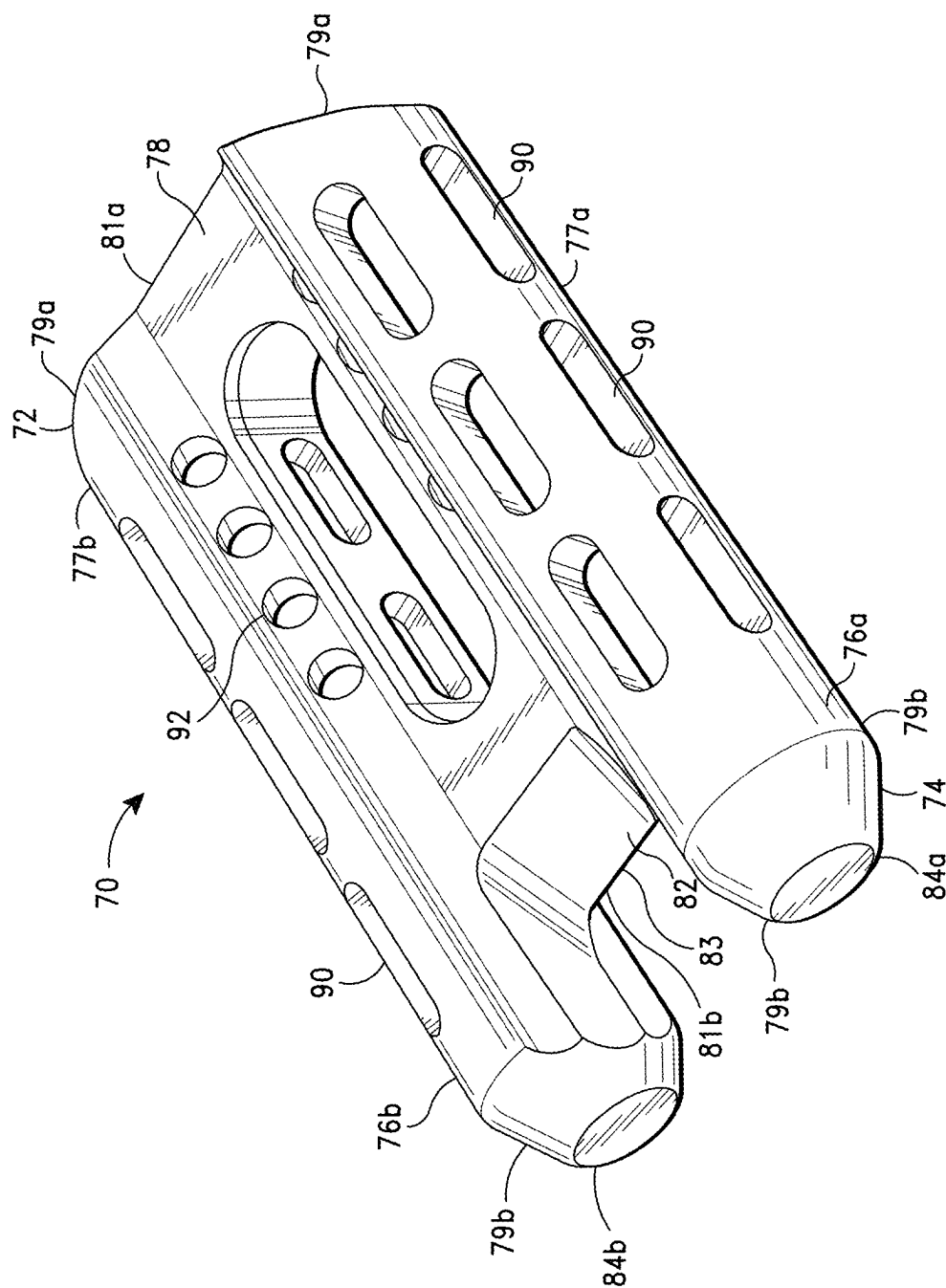


FIG. 6A

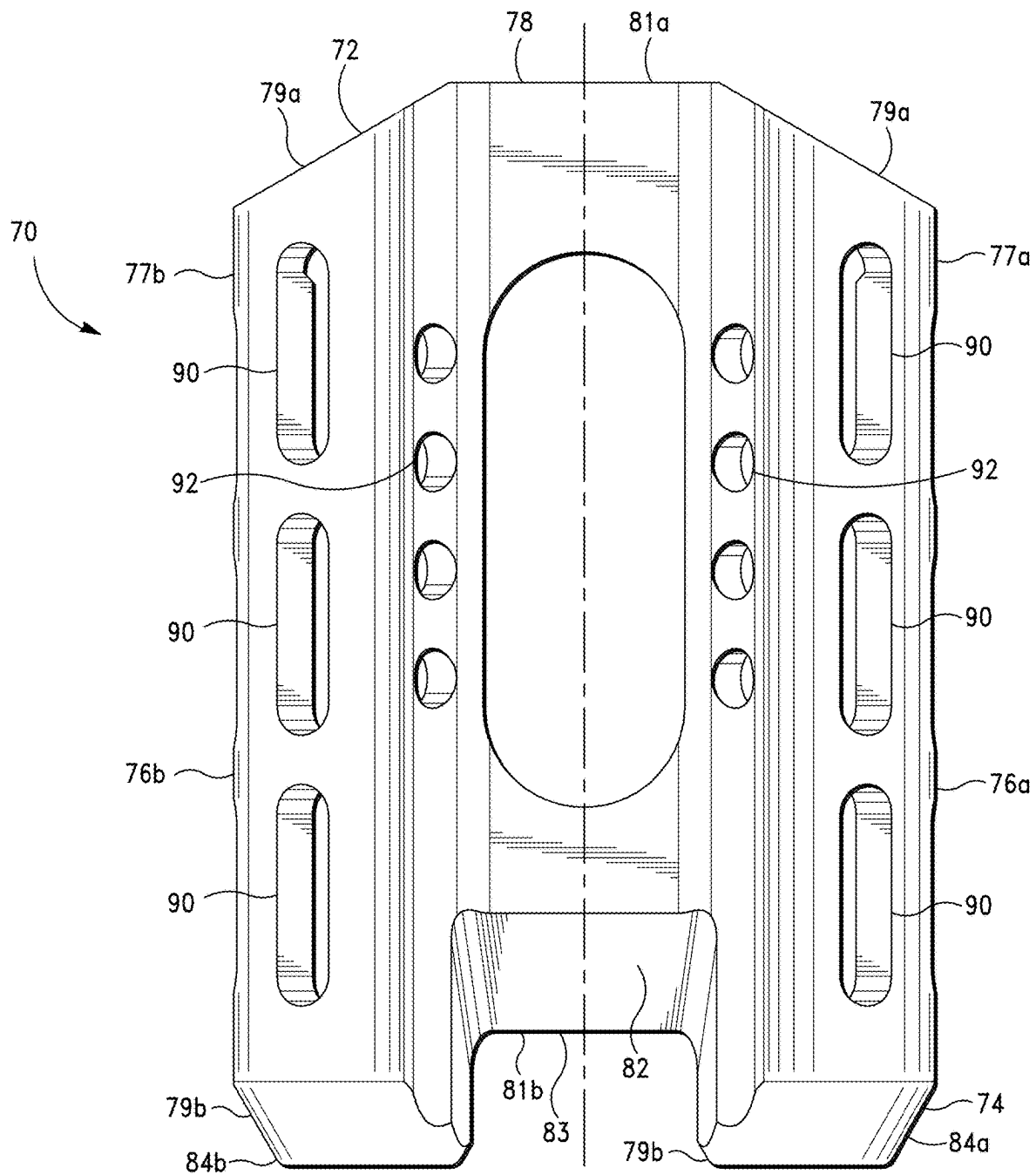


FIG. 6B

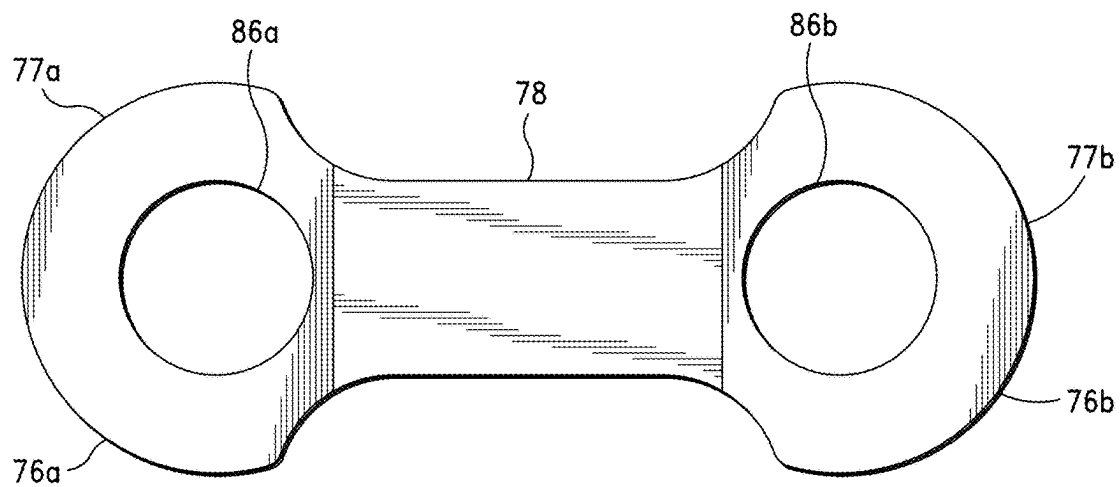


FIG. 6C

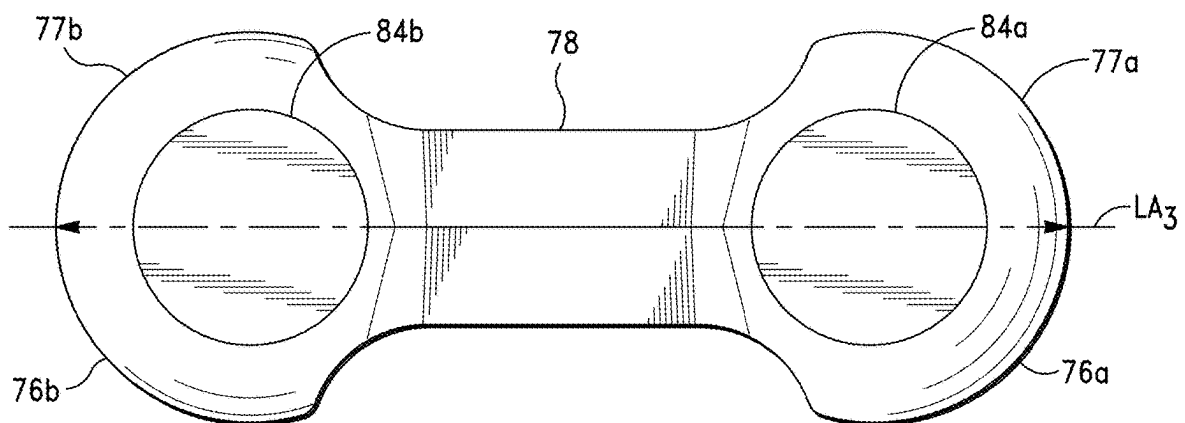
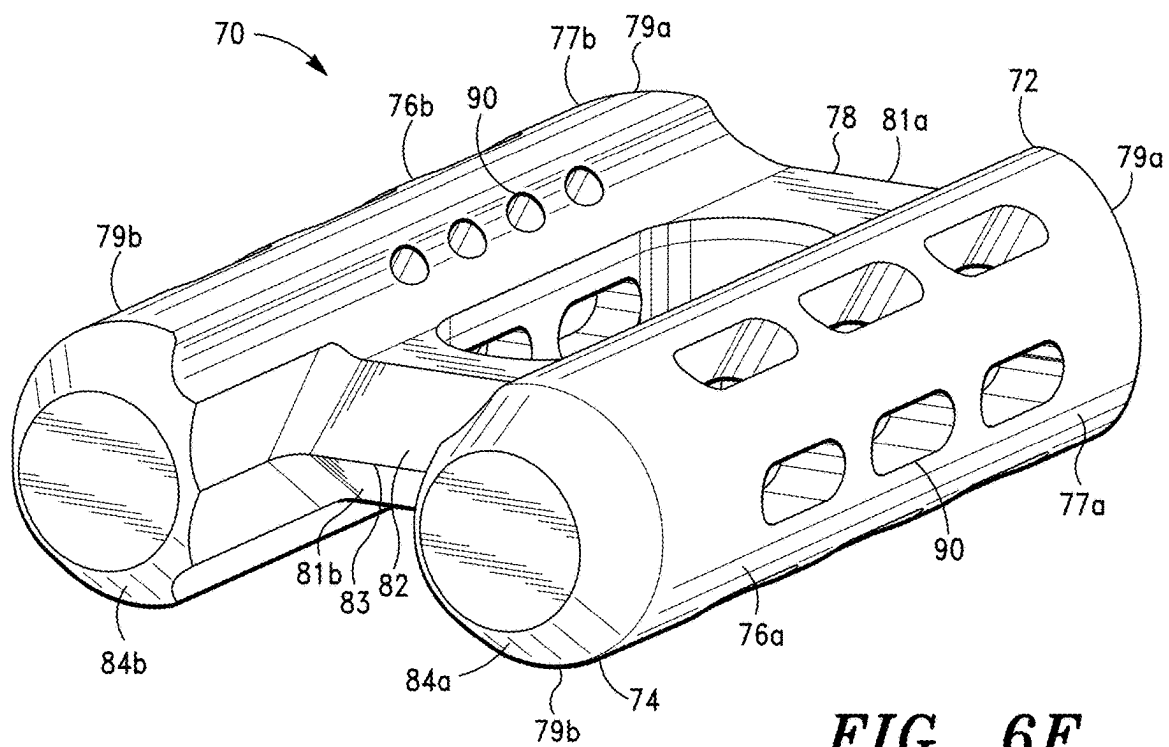
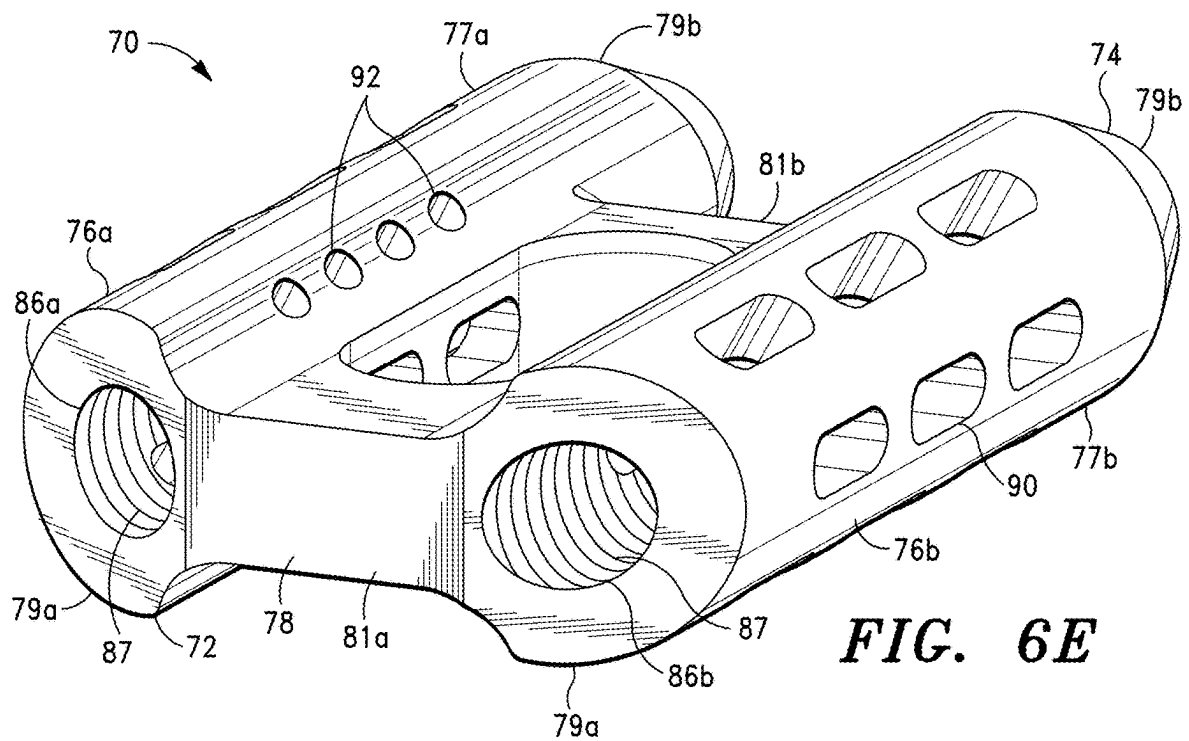


FIG. 6D



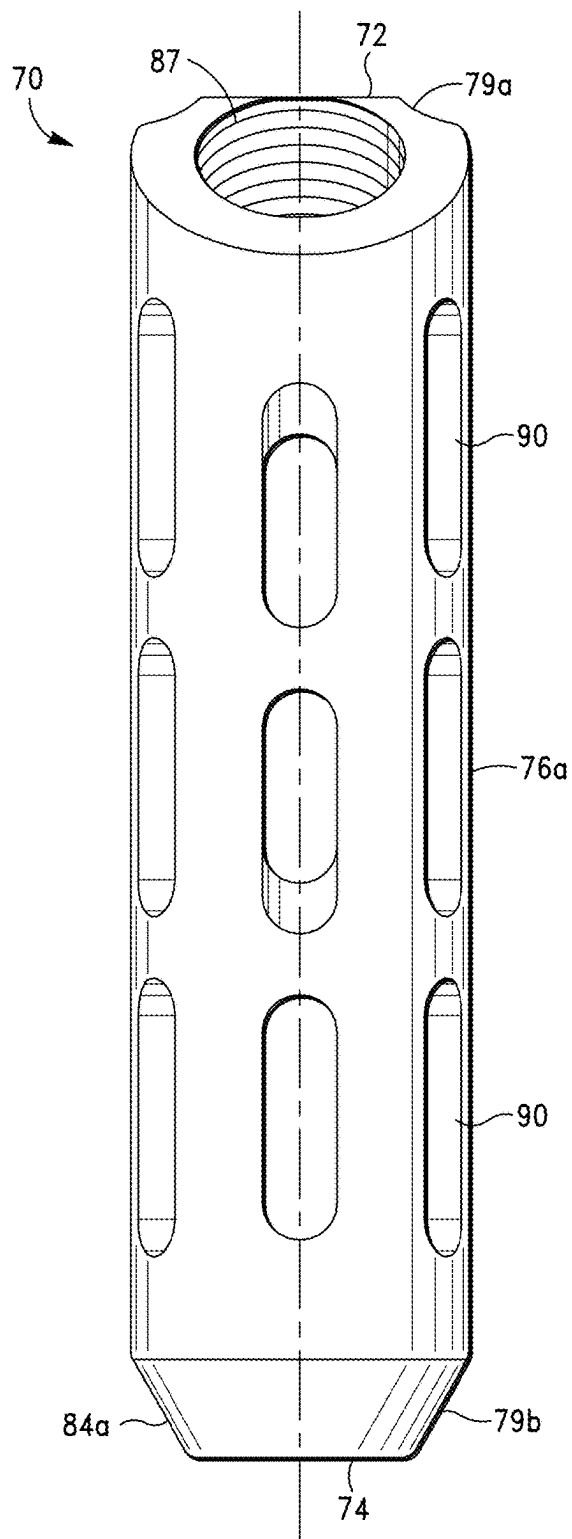


FIG. 6G

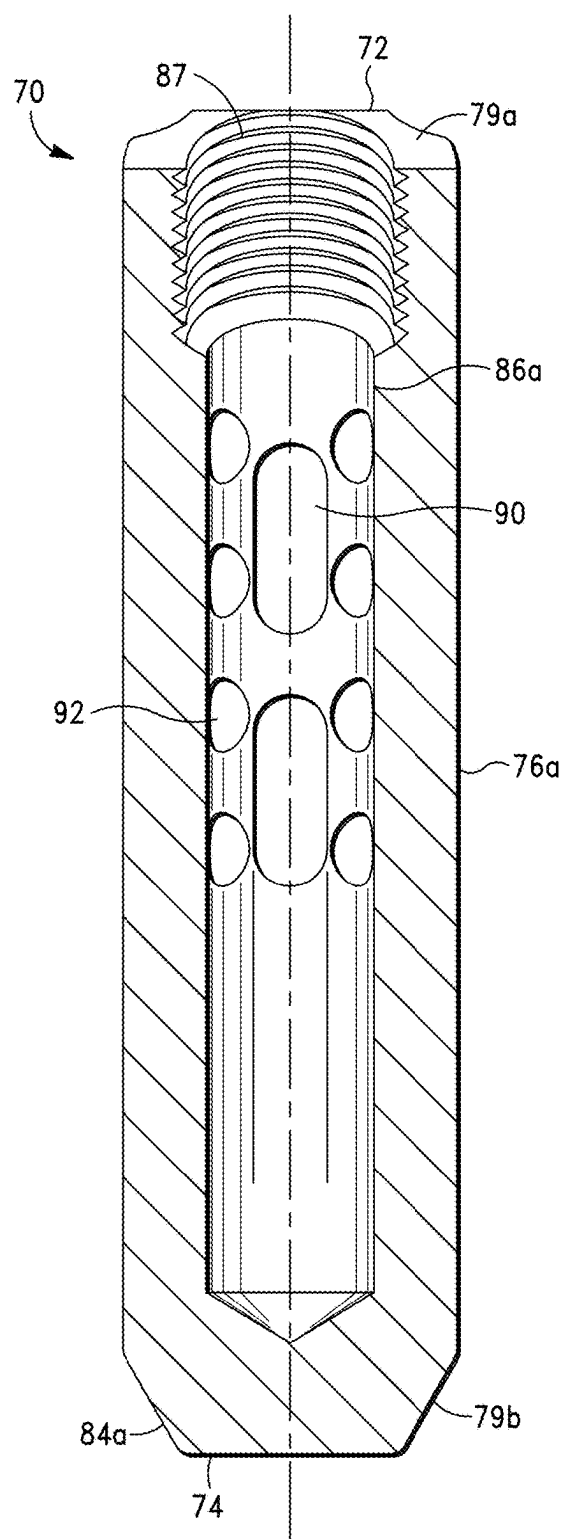
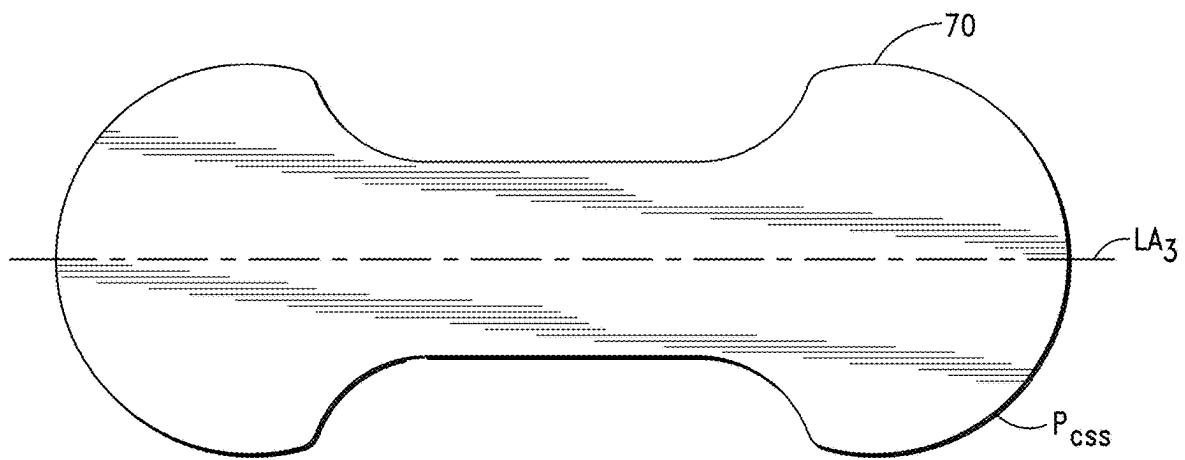


FIG. 6H

*FIG. 6I*

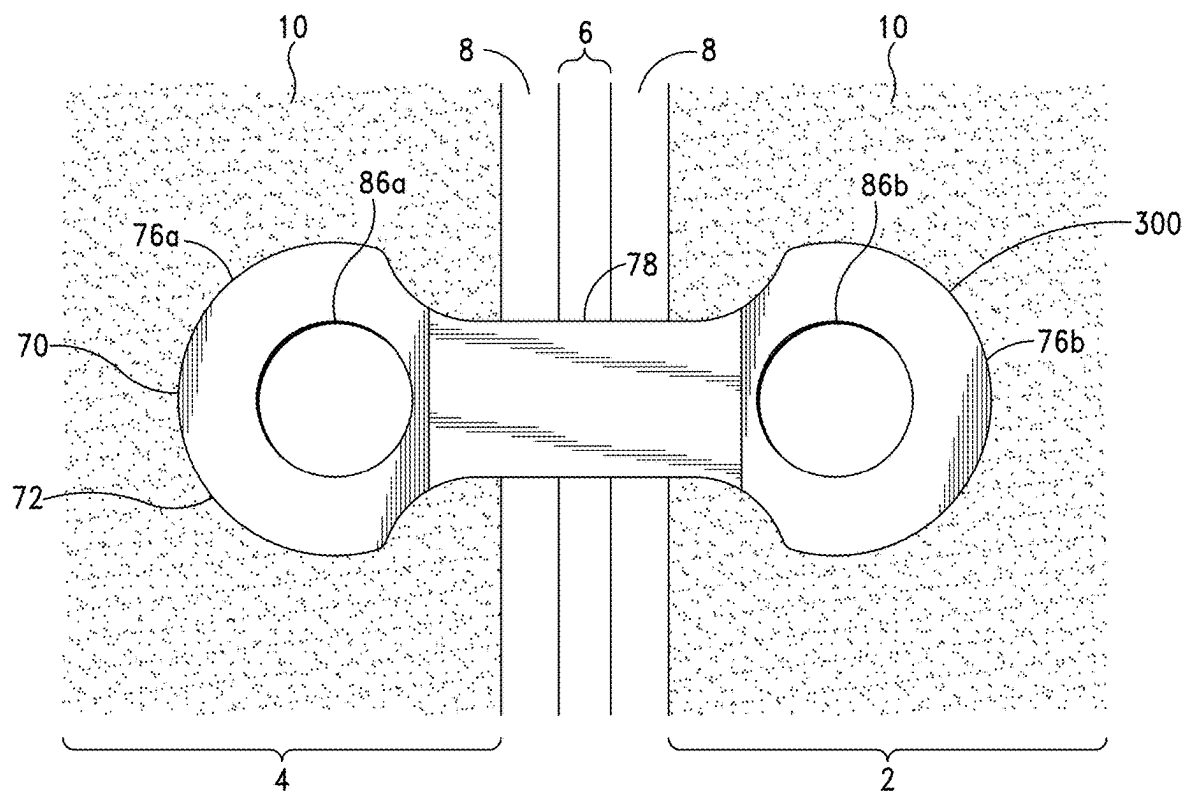


FIG. 7A

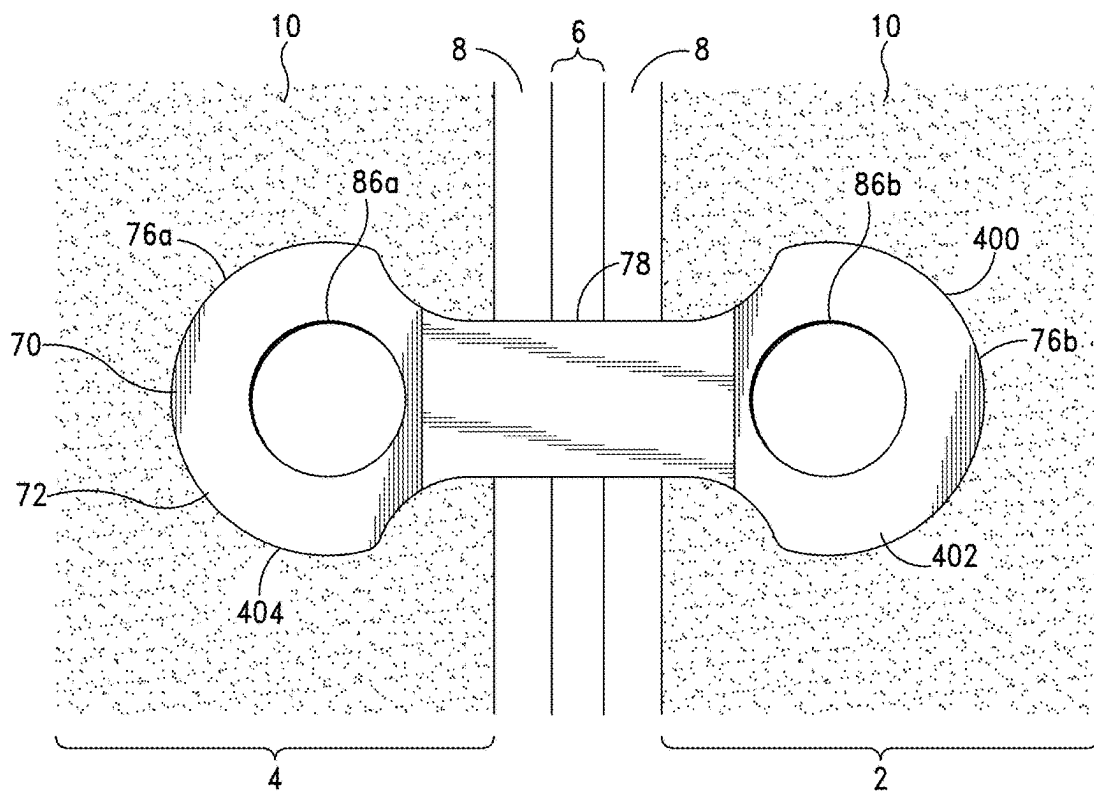


FIG. 7B

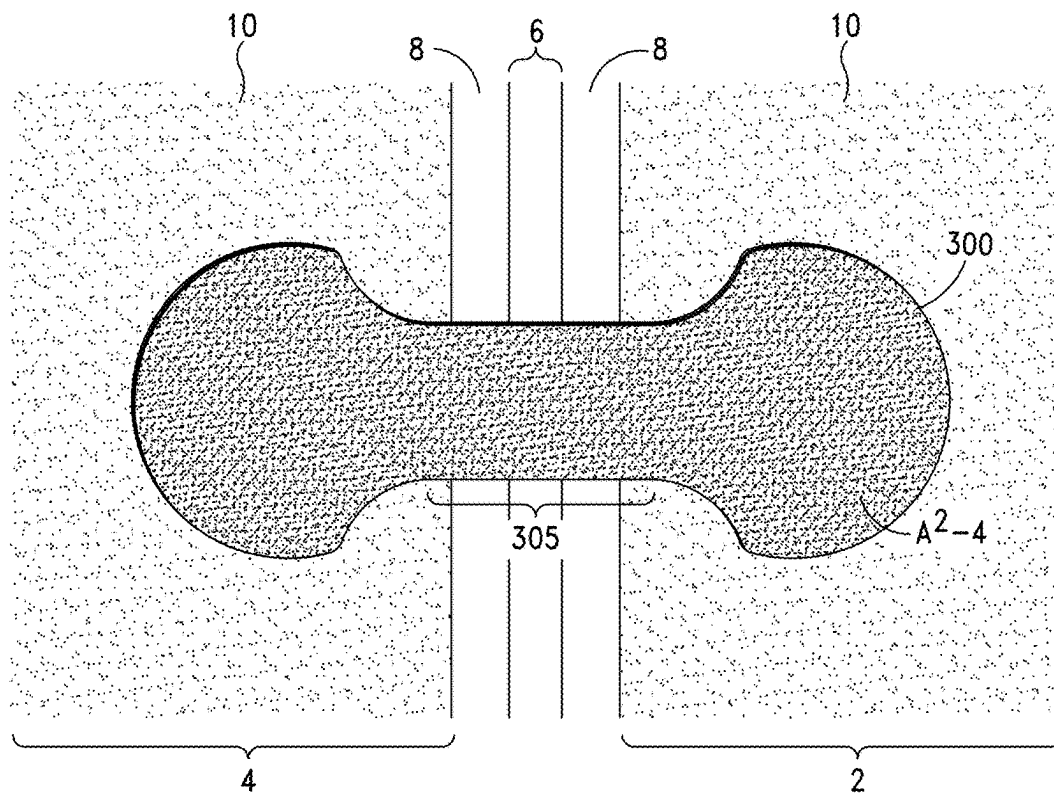


FIG. 8A

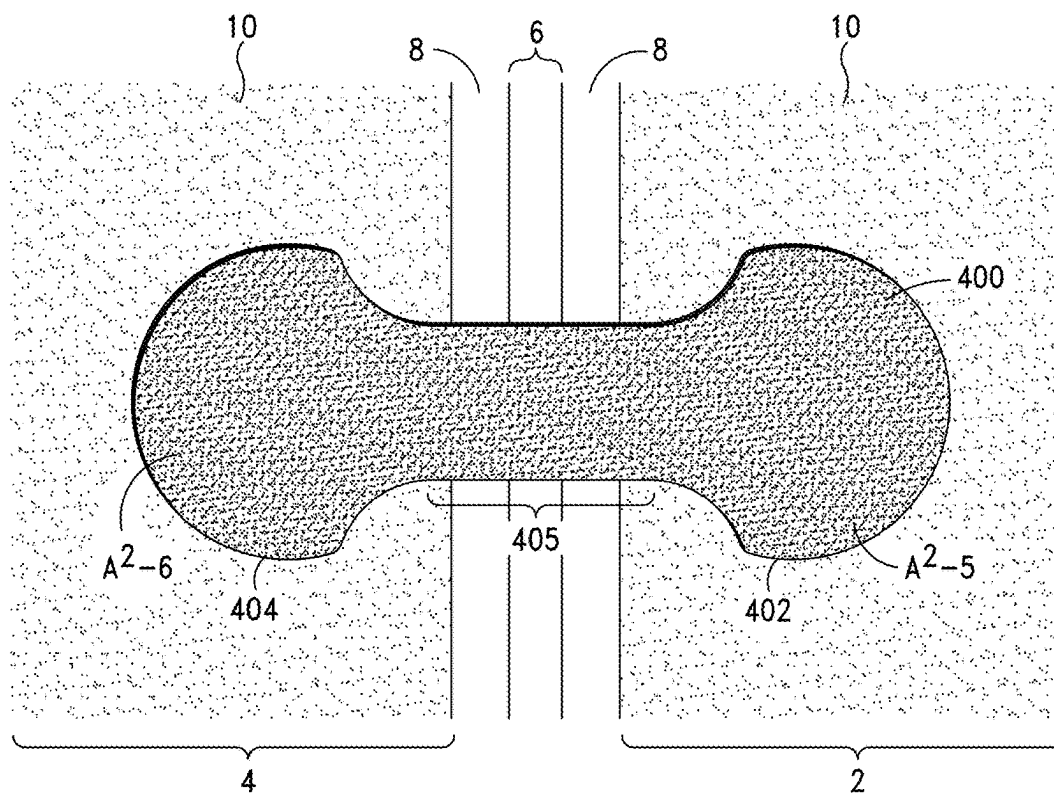


FIG. 8B

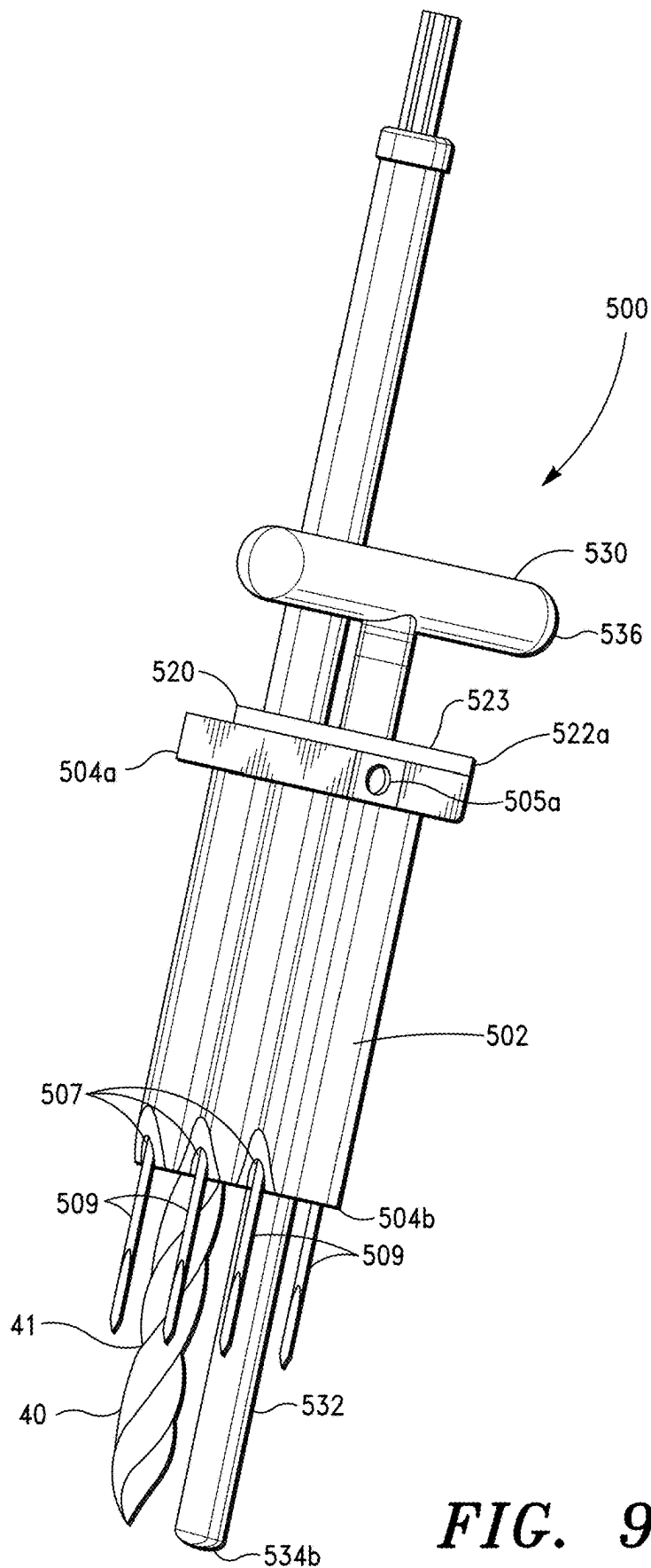


FIG. 9A

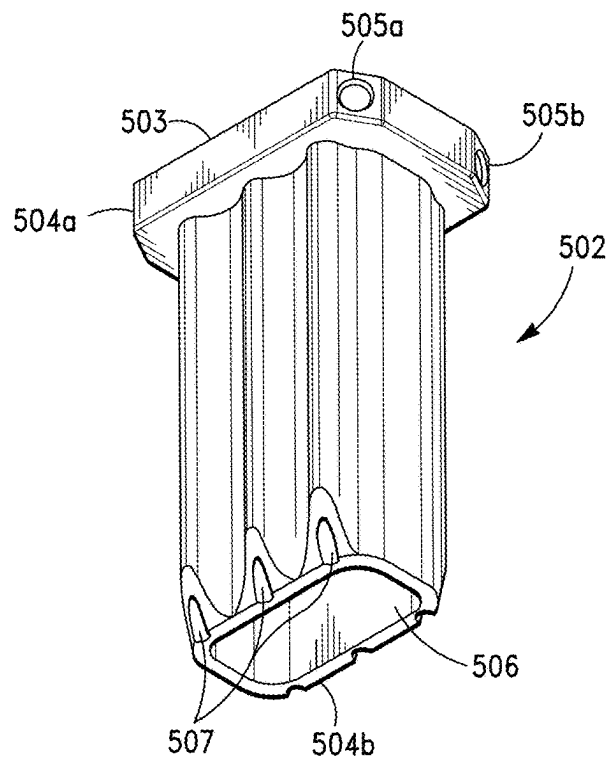


FIG. 9B

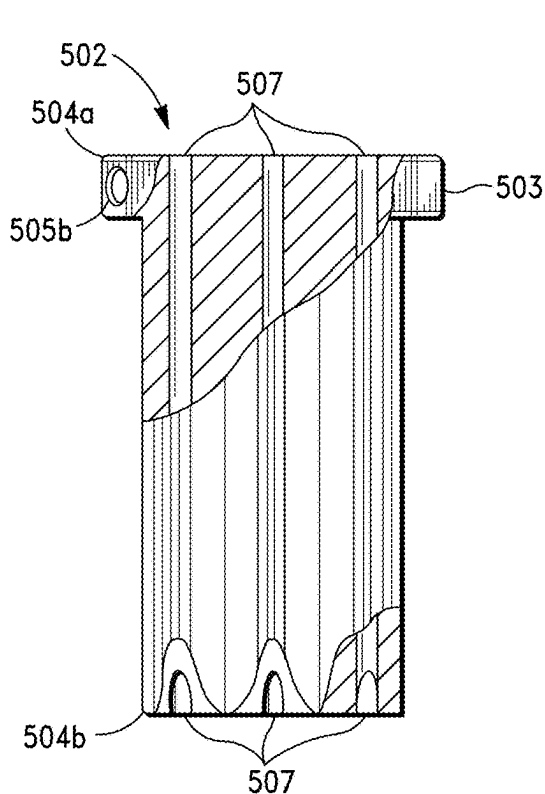


FIG. 9C

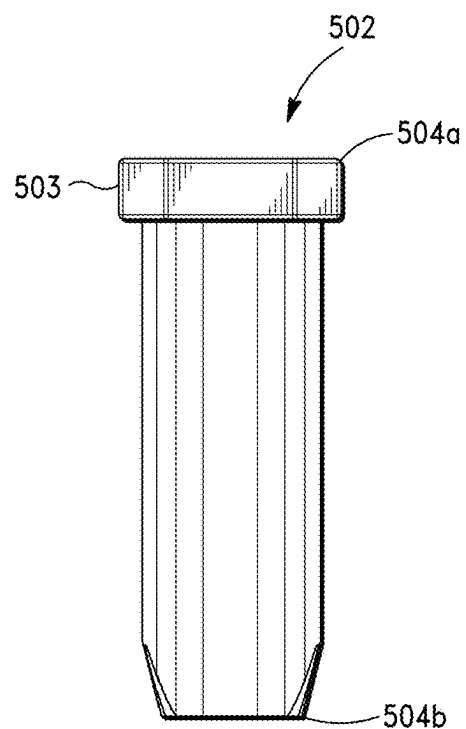


FIG. 9D

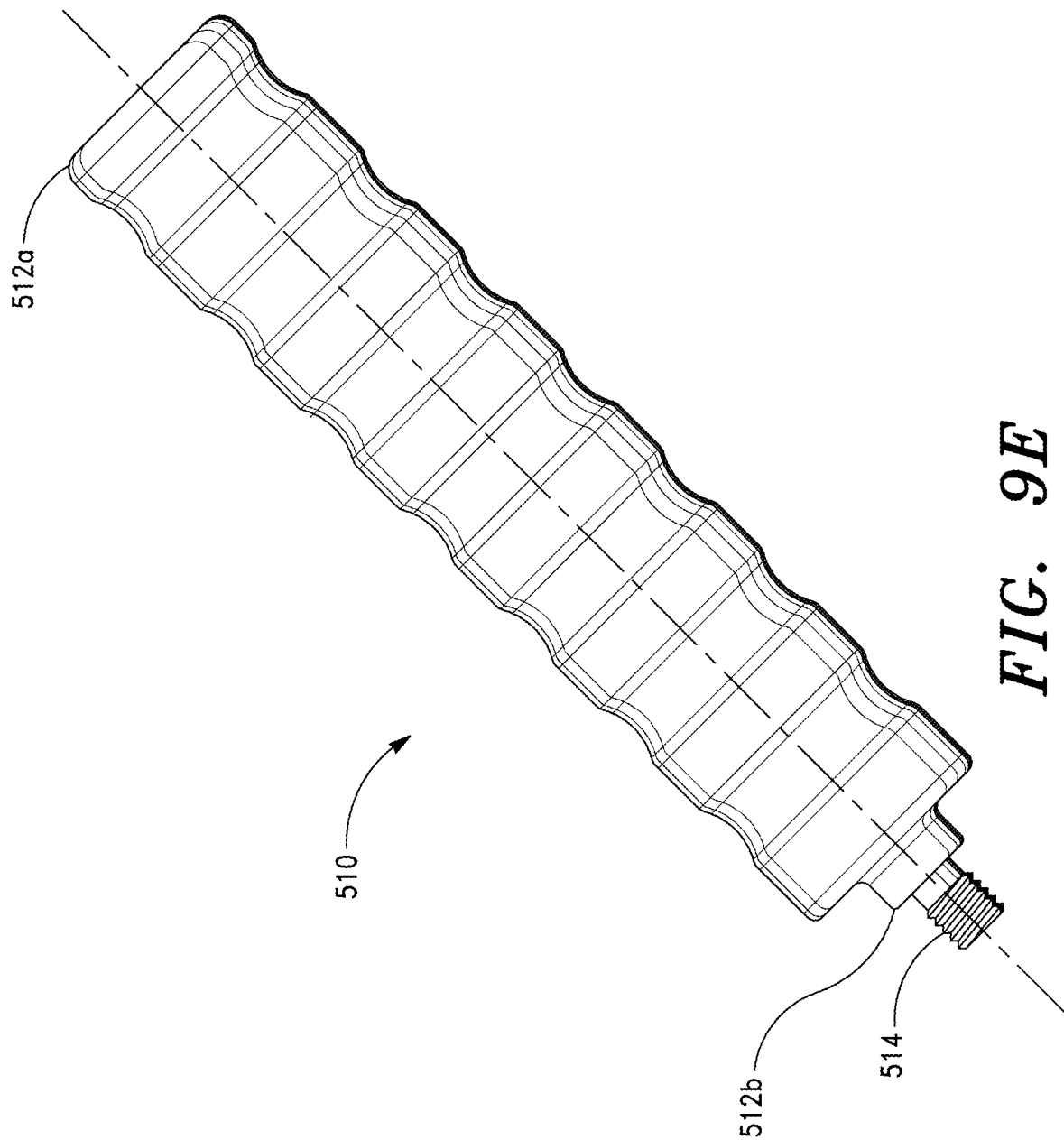


FIG. 9F

FIG. 9E

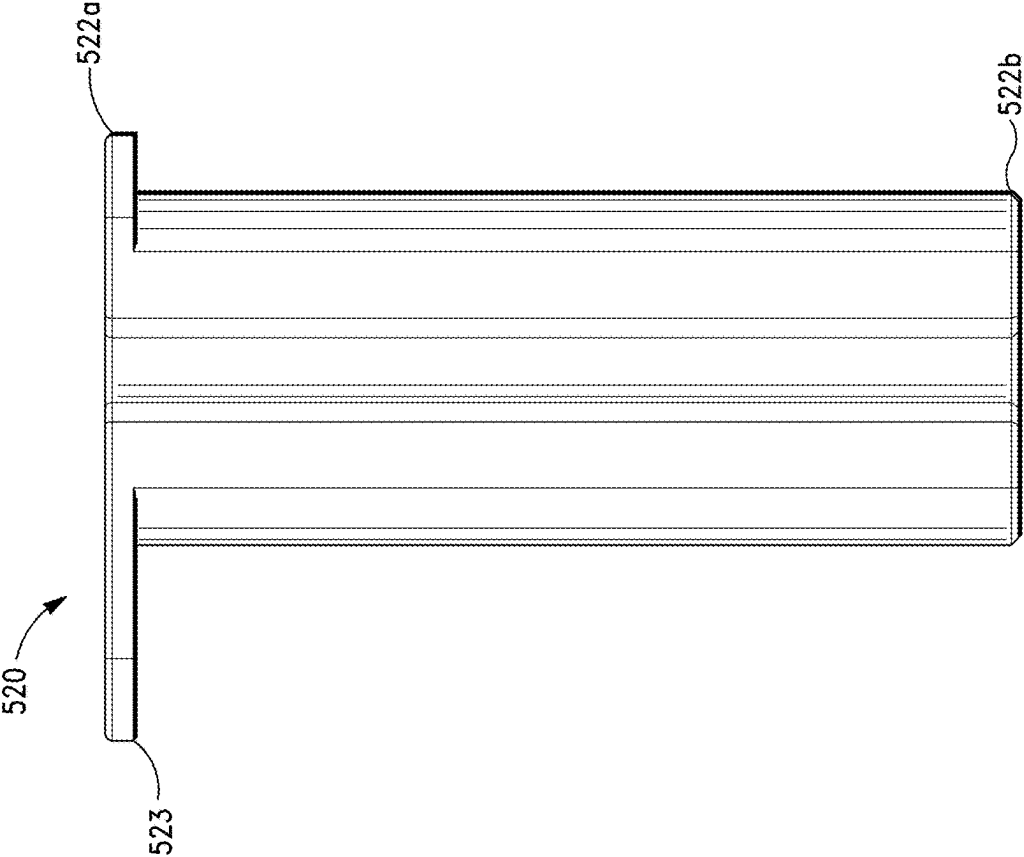


FIG. 9H

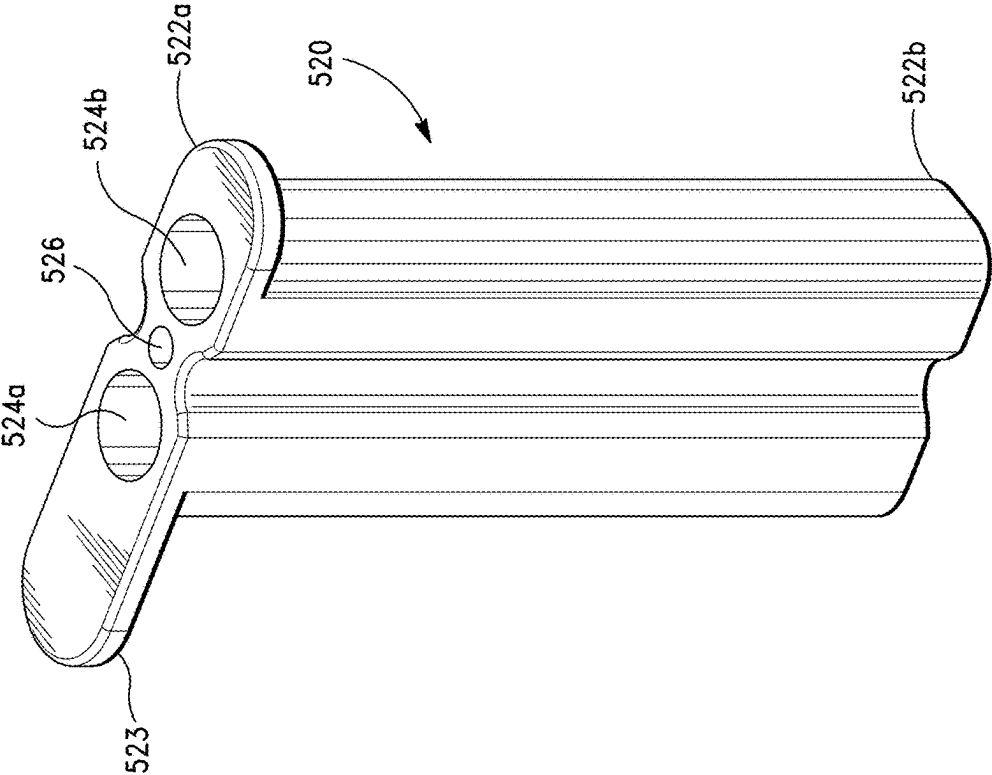


FIG. 9G

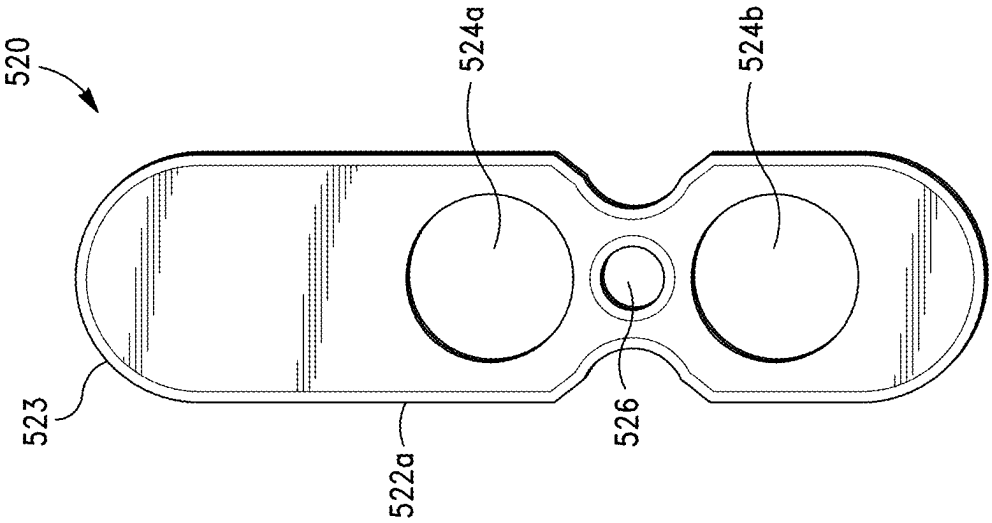


FIG. 9I

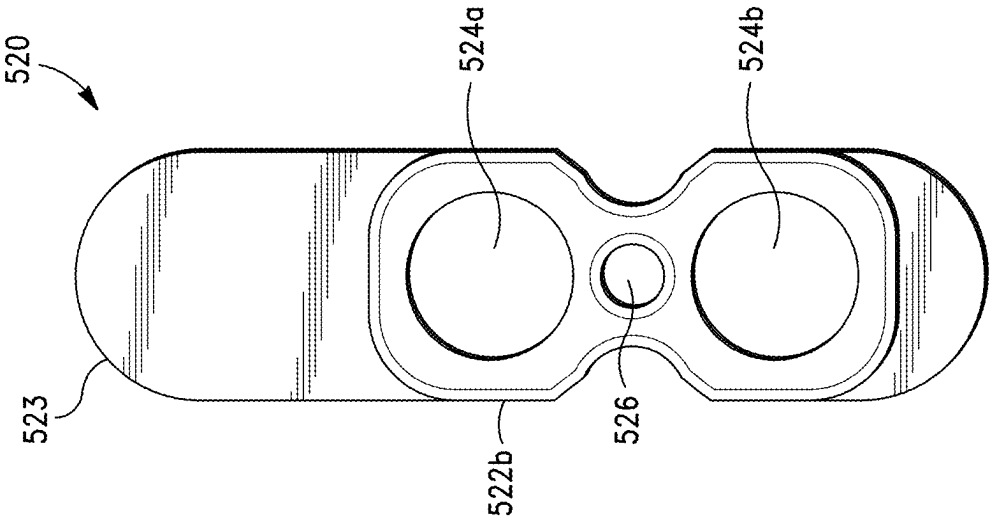
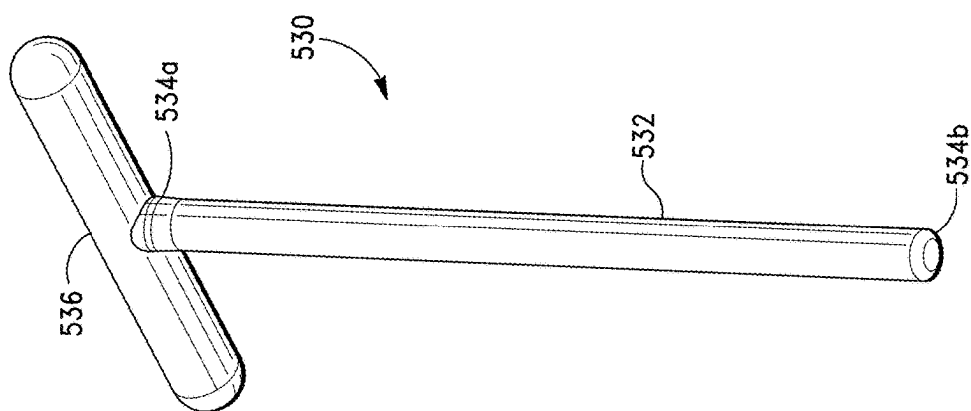
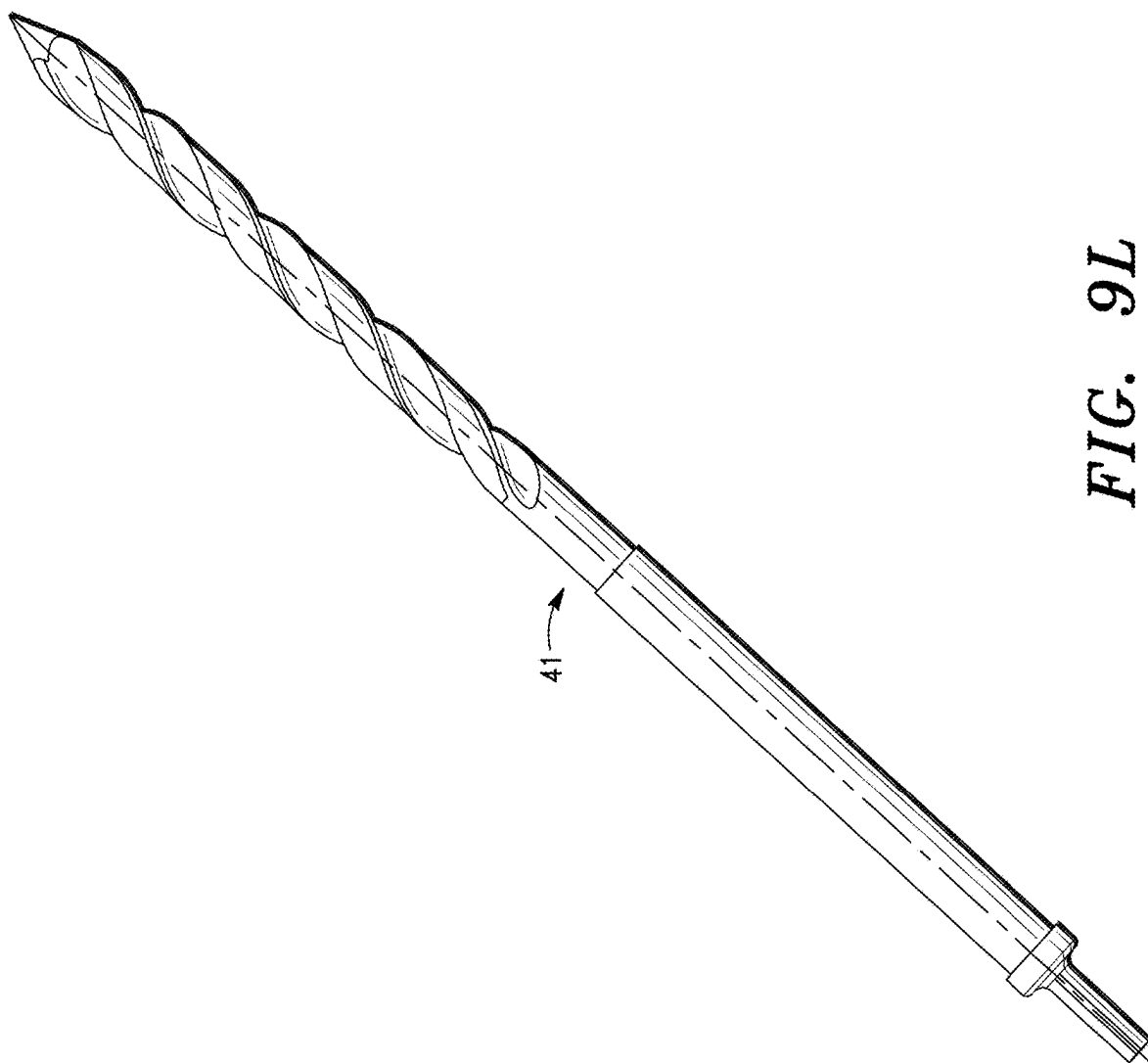
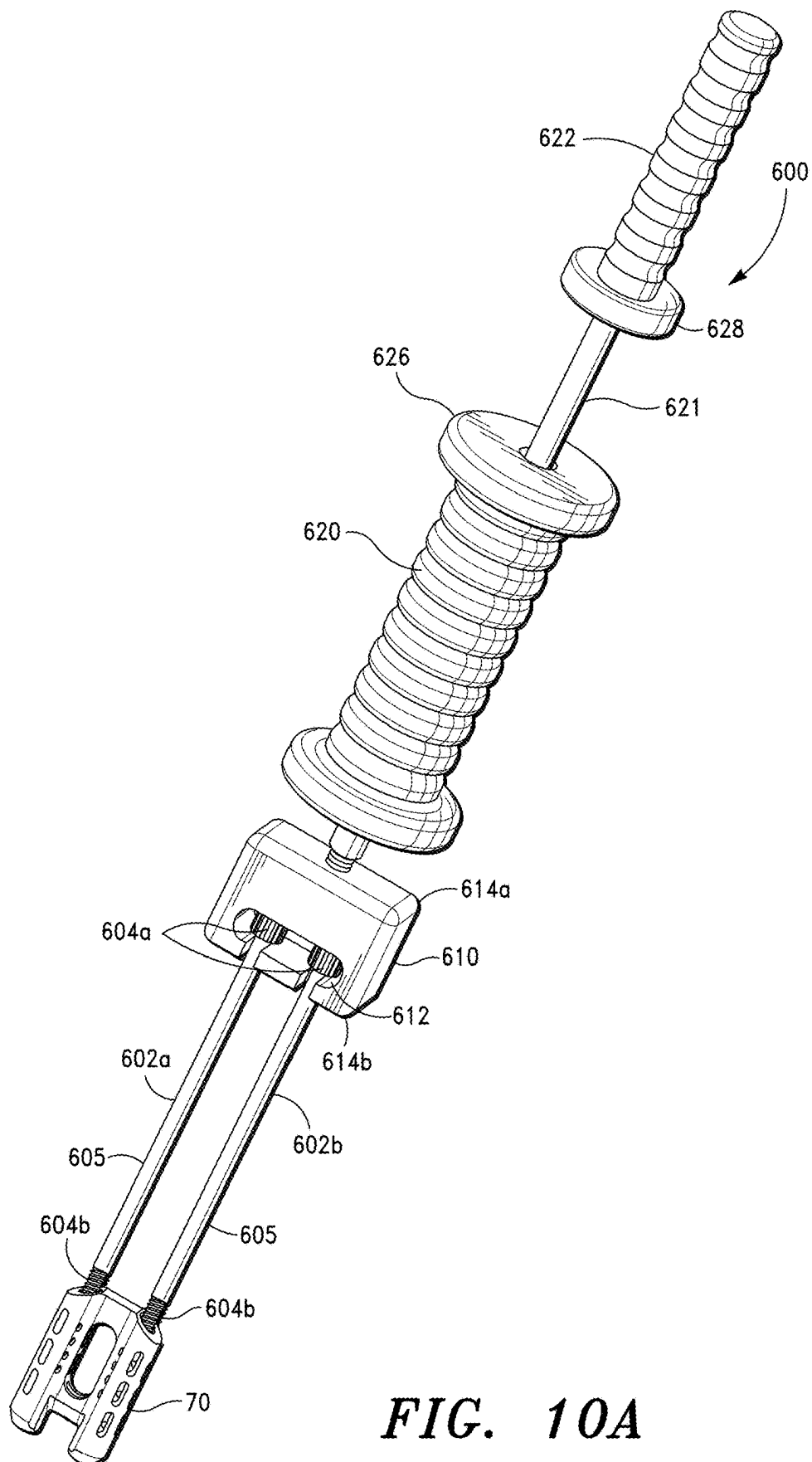


FIG. 9J



**FIG. 10A**

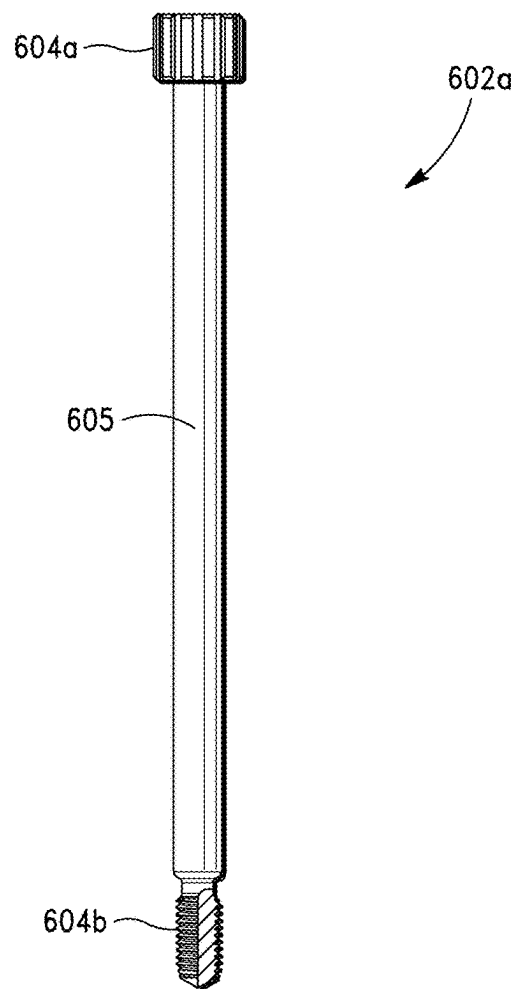


FIG. 10B

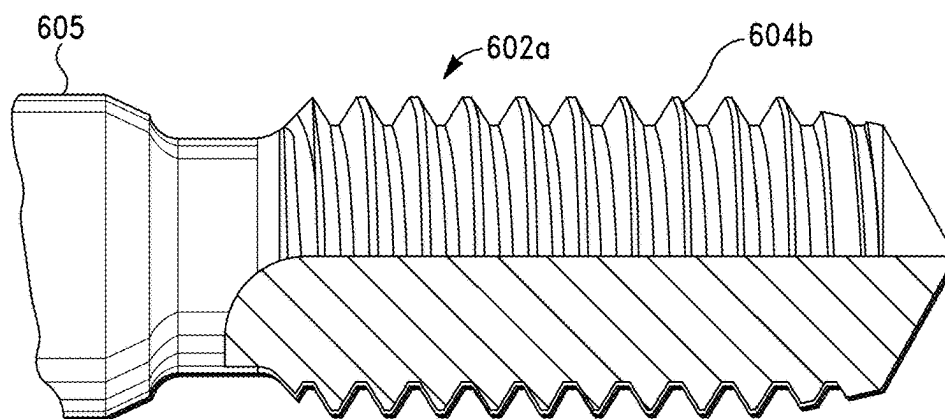


FIG. 10C

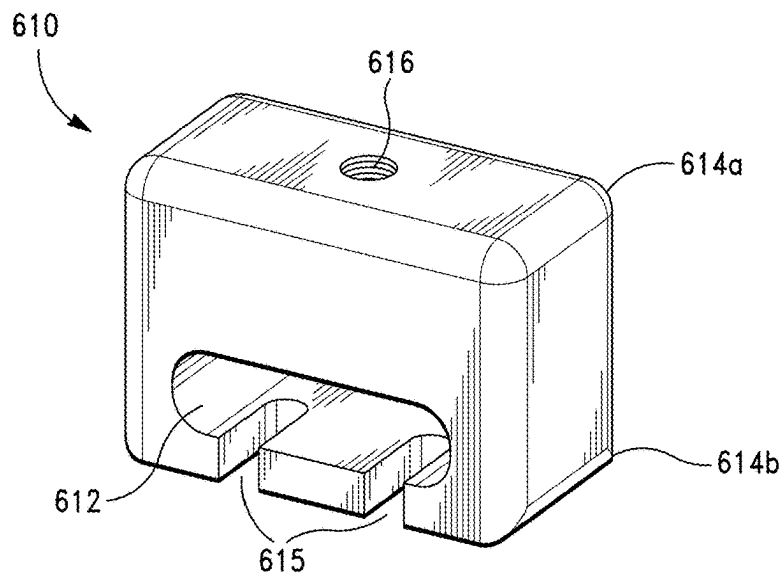


FIG. 10D

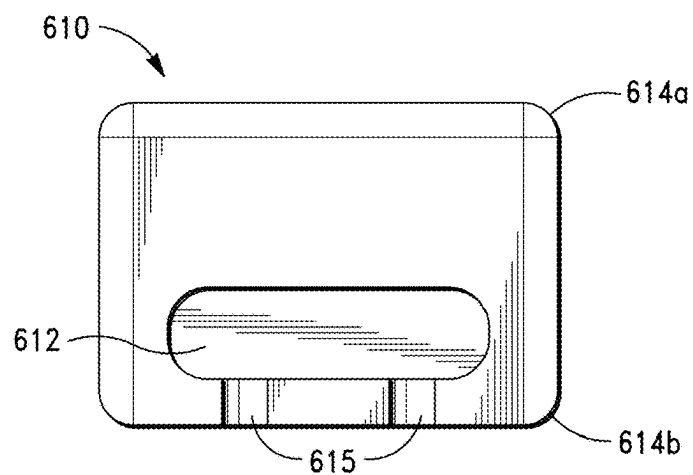


FIG. 10E

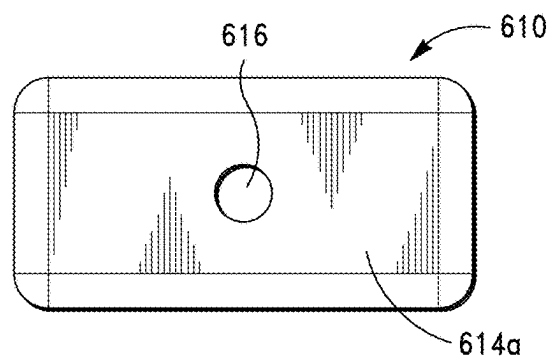


FIG. 10F

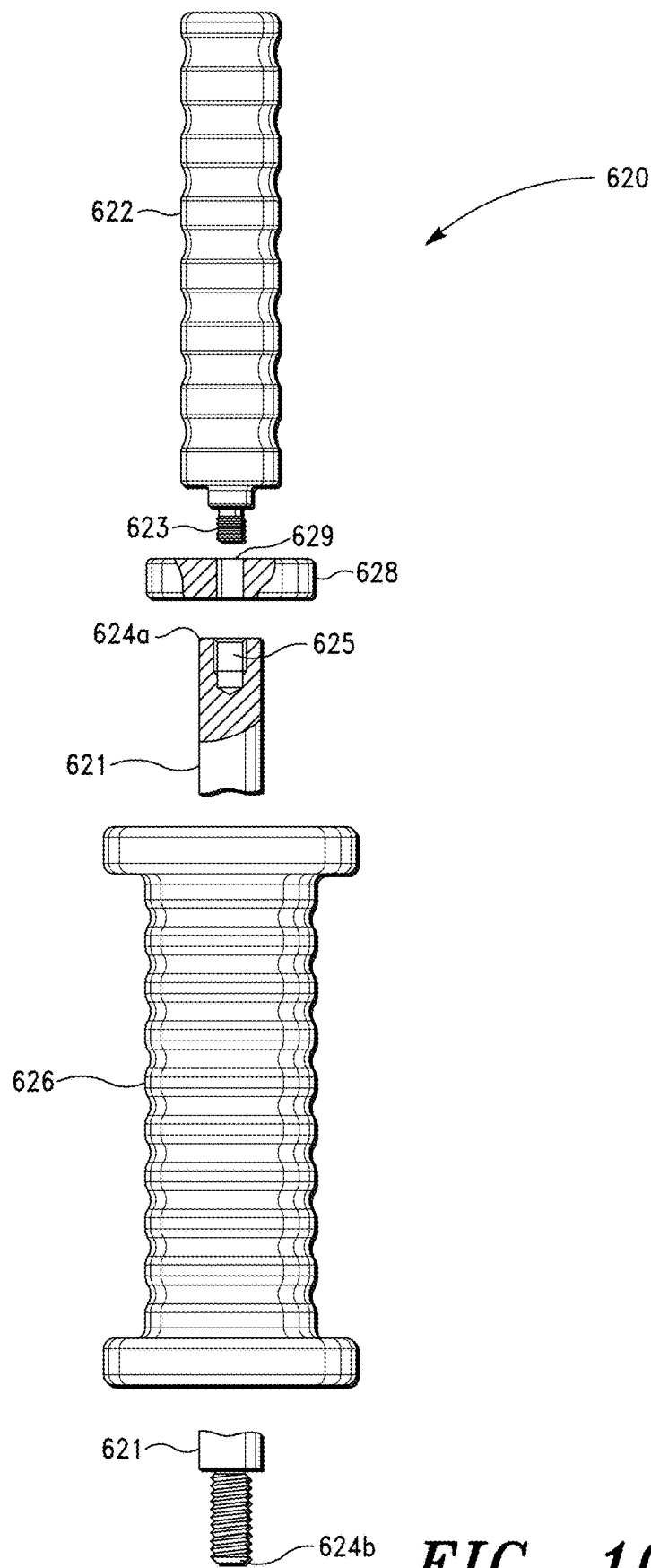


FIG. 10G

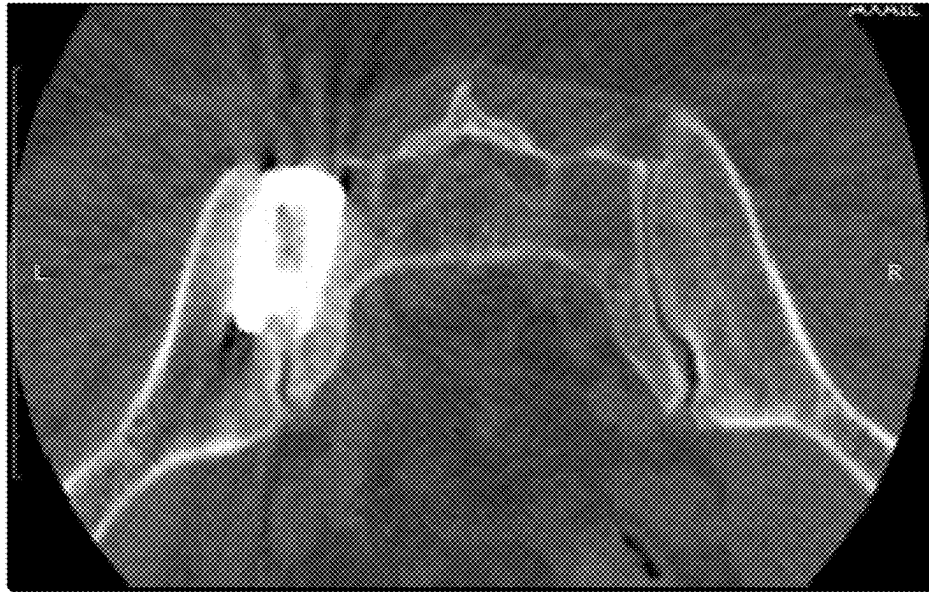


FIG. 11A

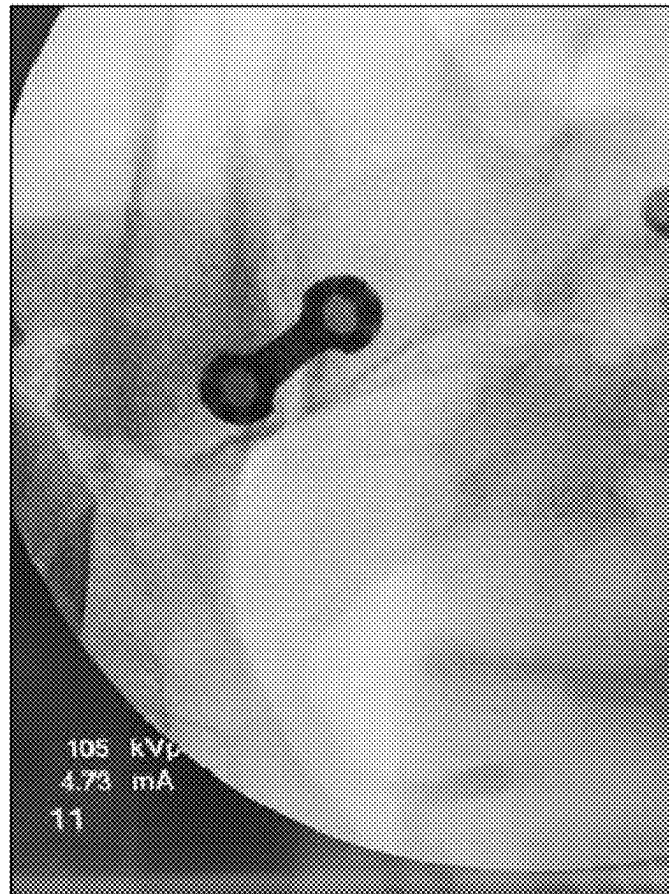
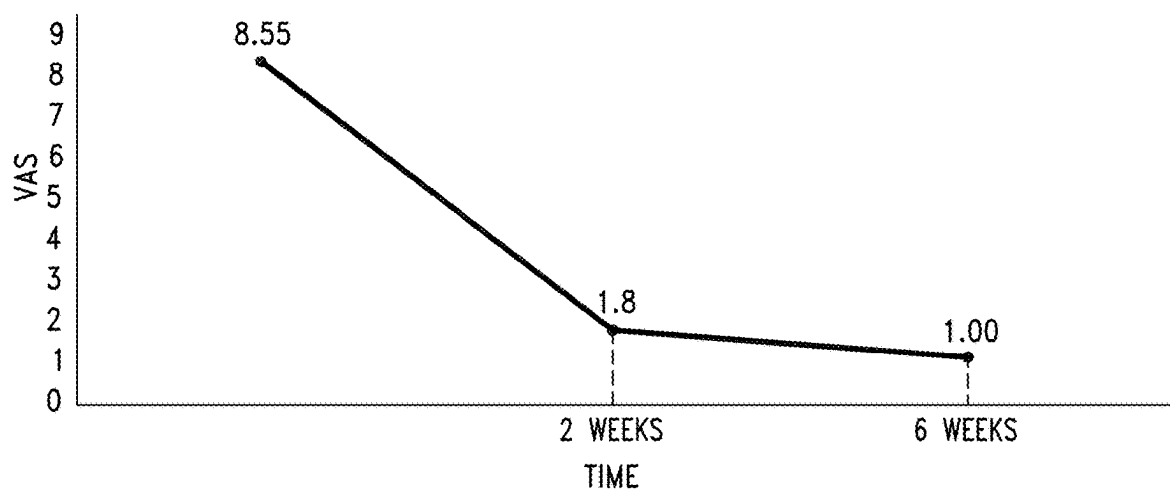


FIG. 11B

*FIG. 12*

1

SYSTEMS FOR SACROILIAC JOINT STABILIZATION

CROSS-REFERENCES TO RELATED APPLICATIONS

This application is a continuation-in-part application of U.S. patent application Ser. No. 13/857,977, filed Apr. 5, 2013, which is a continuation application of U.S. patent application Ser. No. 13/192,289, filed Jul. 27, 2011, now abandoned, which claims the benefit of U.S. provisional patent application Ser. No. 61/368,233, filed Jul. 27, 2010.

FIELD OF THE INVENTION

The present invention relates to methods, systems and apparatus for stabilizing junctions between bone structures. More particularly, the present invention relates to methods, systems and apparatus for stabilizing dysfunctional sacroiliac (SI) joints.

BACKGROUND OF THE INVENTION

As is well known in the art, the sacroiliac (SI) joint 6 comprises a diarthrodial synovial joint, which, as illustrated in FIG. 1A, is defined by the interface between the articular surfaces of the sacrum 2 and the ilium 4. Thus, the SI joint 6 is defined by (and, hence, comprises) portions of the sacrum 2 and ilium 4.

As is also well known in the art, the SI joint further comprises articular cartilage, i.e., hyaline and fibrocartilage, and a strong, extensive ligamentous architecture, which stabilizes the SI joint.

Generally, the articular surfaces of the sacrum 2 and the ilium 4 that define the SI joint 6 comprise cortical bone 8, which is more compact, dense and hard relative to softer trabecular bone 10, which, as further illustrated in FIG. 1A, is disposed in the interior regions of the sacrum and ilium 2, 4.

The SI Joint is distinguished from other synovial joints by the atypical articulation of the different articular surfaces of the sacrum and ilium; the articular surface of the sacrum comprising hyaline cartilage and the articular surface of the ilium comprising substantially stronger fibrocartilage.

As is further well known in the art, the primary plane of motion of the SI joint is anterior-posterior along a transverse axis. The terms often employed to describe the relative motion of the sacrum and ilium are nutation, which refers to anterior-inferior movement of the sacrum while the coccyx (denoted "3" in FIG. 1A) moves posteriorly relative to the ilium, and counternutation, which refers to posterior-superior movement of the sacrum while the coccyx moves anteriorly relative to the ilium.

In most healthy individuals, the SI joint range of motion in flexion-extension is approximately 3°, approximately 1.5° in axial rotation and approximately 0.8° in lateral bending.

As is well established, the SI joint performs several seminal biomechanical functions. The primary functions of the SI joint are to attenuate loads exerted on the upper body and to distribute the loads to the lower extremities. The SI joint also functions as a shock absorber for loads exerted on spine.

As is also well established, the noted loads and, hence, forces exerted on the SI joint can adversely affect the biomechanical functions of the SI joint, which can, and often will, result in SI joint dysfunction—an often-overlooked musculoskeletal pathology associated with lower back pain.

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Indeed, SI joint dysfunction is estimated to be the primary cause of lower back pain in 15-30% of subjects afflicted with such pain. However, lower back pain associated with SI joint dysfunction is suspected to be far more common than most healthcare providers realize, since such pain is often associated with other skeletal and musculoskeletal dysfunctions.

SI joint dysfunction, and pain associated therewith, can be caused by various SI joint abnormalities and/or disorders, including traumatic fracture dislocation of the pelvis, degenerative arthritis, sacroiliitis, i.e., an inflammation or degenerative condition of the sacroiliac joint; osteitis condensans ilii, and other degenerative conditions of the SI joint structures.

Various non-surgical methods, such as administration of pharmacological agents, e.g., the corticosteroid prednisone, and surgical methods and devices, i.e., prostheses, have been developed and employed to treat SI joint dysfunction.

The most common approach employed to treat SI joint dysfunctions (when non-surgical treatments fail to ameliorate pain associated therewith), at present, is SI joint stabilization, i.e., reinforcing or modulating articulation by and between the sacrum and ilium, via surgical intervention.

SI joint stabilization typically comprises surgical placement of a prosthesis proximate to or in a dysfunctional SI joint and is generally characterized by the direction of access to the dysfunctional SI joint, i.e., anterior, posterior or lateral.

Although several conventional SI joint stabilization surgical methods and associated bone prostheses have effectively ameliorated pain associated with SI joint dysfunction, there remains many disadvantages associated with the conventional methods and associated prostheses.

A major disadvantage associated with many conventional SI joint stabilization surgical methods is that the surgeon is required to make a substantial incision in and through the skin and tissues of a subject to access the dysfunctional SI joint. Often referred to as "open surgery" methods, these surgical methods have the attendant disadvantages of requiring general anesthesia and often involve increased operative time, pain, hospitalization, and recovery time due to the extensive soft tissue damage. There is also an increased probability of post-surgical complication associated with open surgery methods, such as nosocomial infection.

Minimally-invasive methods for SI joint stabilization have thus been developed to address the noted disadvantages associated with open surgery methods. Although conventional minimally-invasive SI joint stabilization methods, such as the methods disclosed in U.S. Pub. No. 2009/0076551 to Petersen, have garnered some success in relieving pain associated with SI joint dysfunction and have effectively addressed many of the disadvantages associated with open surgery methods, there similarly remains many disadvantages associated with conventional minimally-invasive SI joint stabilization methods.

A major disadvantage associated with many conventional minimally-invasive SI joint stabilization methods is that such methods are difficult to perform and, hence, often require extensive, system-specific surgical training and experience. Despite the level of surgical training and experience that surgeons possess, when such conventional minimally-invasive SI joint stabilization methods are employed, there is still a substantial incidence of damage to the lumbosacral neurovascular structures proximate to the SI joint.

A further disadvantage associated with many conventional minimally-invasive SI joint stabilization methods and associated apparatus, i.e., prostheses, such as the methods

and prostheses disclosed in U.S. Pub. No. 2009/0076551 to Petersen, is that pre-existing sacral abnormalities can lead to displacement of the implanted prostheses, which can, and often will result in damage to surrounding bone and soft tissue structures.

An additional disadvantage associated with many conventional minimally invasive SI joint stabilization methods is that they comprise anterior or lateral approaches to the dysfunctional SI joint and, hence, muscles, e.g., gluteal aponeurotic fascia and gluteus medius, and ligaments are typically disrupted, and nerves and blood vessels are susceptible to damage during placement of a prosthesis in a dysfunctional SI joint.

Further, some conventional minimally-invasive SI joint stabilization methods are particularly prone to failure due to displacement of the prostheses in the dysfunctional SI joint and/or failure of the prostheses to effectively engage the SI joint structures, e.g., articular surfaces of the sacrum and/or ilium.

Various "improved" prostheses have thus been developed for use in minimally-invasive SI joint stabilization methods or procedures to effectively engage SI joint structures and maintain engagement thereto during SI joint function.

Although many of the "improved" prostheses, when deployed properly in a dysfunctional SI joint, can, and often will, effectively engage SI joint structures, there remains several disadvantages associated with the prostheses. Illustrative are the prostheses disclosed in U.S. Pat. No. 8,951,254 to Mayer, et al.

The prostheses disclosed in U.S. Pat. No. 8,951,254 comprise or are coated with a liquefiable synthetic polymer that is adapted to liquify upon administration of mechanical energy, e.g., high frequency vibration, when implanted and re-solidify thereafter to securely engage the SI joint structures, i.e., sacrum and ilium.

A major disadvantage associated with the prostheses disclosed in U.S. Pat. No. 8,951,254 is that the liquefiable synthetic polymers, when re-solidified in situ, are structurally inferior to the osseous or bone tissue of the sacrum and ilium. The fusion sites between the articular surfaces of the sacrum and ilium that define the SI joint are, thus, highly susceptible to structural fatigue and failure, which can, and often will, result in misalignment of the SI joint and ultimately increased pain for the subject.

A further disadvantage associated with the prostheses disclosed in U.S. Pat. No. 8,951,254 is that the synthetic liquefiable synthetic polymers are also substantially immunogenic and will induce an adverse immune response when the prostheses are implanted in a dysfunctional SI joint. As is well established, the adverse immune response can, and often will, prevent healing and osteogenic processes, e.g., remodeling of damaged osseous tissue and regeneration of new osseous tissue.

Additional disadvantages associated with the prostheses disclosed in U.S. Pat. No. 8,951,254 and many other prostheses designed for minimally-invasive SI joint stabilization are that the noted prostheses are difficult to accurately place in optimum positions in a dysfunctional SI joint and, in many instances, lack sufficient structural properties, such as rigidity and/or fatigue resistance, to effectively stabilize the dysfunctional SI joint.

It would thus be desirable to provide SI joint stabilization systems and apparatus, which substantially reduce or eliminate the disadvantages associated with conventional SI joint stabilization systems and apparatus.

It is therefore an object of the invention to provide improved SI joint stabilization systems and apparatus, which

substantially reduce or eliminate the disadvantages associated with conventional SI joint stabilization systems and apparatus.

It is another object of the invention to provide improved minimally-invasive SI joint stabilization systems and apparatus, which can be readily employed to place prostheses in and, thereby, stabilize dysfunctional SI joints via a posterior approach.

It is another object of the invention to provide improved minimally-invasive SI joint stabilization systems and apparatus, which can be readily employed to stabilize dysfunctional SI joints.

It is another object of the invention to provide improved minimally-invasive SI joint stabilization systems and apparatus, which, when implanted in a dysfunctional SI joint, effectively ameliorate pain associated with the SI joint dysfunction.

It is another object of the invention to provide improved SI joint prostheses that can readily be employed in minimally-invasive SI joint stabilization methods and provide secure engagement to SI joint structures.

It is another object of the invention to provide improved SI joint prostheses that can readily be employed in minimally-invasive SI joint stabilization methods and possess optimal structural properties to effectively stabilize dysfunctional SI joints.

It is yet another object of the invention to provide improved SI joint prostheses that can readily be employed in minimally-invasive SI joint stabilization methods and facilitate remodeling of damaged osseous tissue and regeneration of new osseous tissue and osseous tissue structures.

SUMMARY OF THE INVENTION

The present invention is directed to minimally-invasive methods, systems and apparatus for stabilizing dysfunctional SI joints.

In some embodiments of the invention, there are thus provided minimally-invasive systems for stabilizing dysfunctional SI joints. In one embodiment, the minimally-invasive system generally comprises:

a tool assembly adapted to access a dysfunctional SI joint of a subject via a posterior approach, the tool assembly comprising an elongated guide probe, an SI joint opening or defect creation assembly, and prosthesis deployment assembly,

the elongated guide probe being adapted to be positioned in the dysfunctional SI joint and function as a guide for advancing the defect creation assembly into the dysfunctional SI joint and placement of a prosthesis therein,

the defect creation assembly being adapted to create a pilot SI joint opening in the dysfunctional SI joint, the defect creation assembly comprising a guide member lumen that extends through the defect creation assembly, the guide member lumen being adapted to receive the guide probe therein to facilitate translation of the defect creation assembly thereon,

the defect creation assembly further comprising a bone dislodging member adapted to dislodge portions of bone in the dysfunctional SI joint,

the pilot SI joint opening comprising a first portion disposed in the sacrum and comprising a first sacrum opening cross-sectional shape and a second portion disposed in the ilium and comprising a first ilium opening cross-sectional shape,

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the prosthesis deployment assembly comprising prosthesis engagement means; and

a prosthesis comprising first and second elongated partially cylindrical sections connected to a bridge section, the first partially cylindrical surface region comprising a first partially cylindrical surface region cross-sectional shape that corresponds to at least a first portion of the first sacrum opening cross-sectional shape, the second partially cylindrical surface region comprising a second partially cylindrical surface region cross-sectional shape that corresponds to at least a first portion of the first ilium opening cross-sectional shape, the bridge section distal end comprising a first tapered region configured and adapted to disrupt at least articular cartilage and cortical bone,

the first elongated partially cylindrical section of the prosthesis comprising a first internal prosthesis engagement member lumen and the second elongated partially cylindrical section of the prosthesis comprising a second internal prosthesis engagement member lumen,

the first and second internal prosthesis engagement member lumens adapted to cooperate with the prosthesis engagement means of the prosthesis deployment assembly,

the prosthesis further comprising a plurality of slots in communication with the first and second internal prosthesis engagement member lumens,

the prosthesis being sized and adapted to be press-fit into the pilot SI joint opening, wherein said prosthesis induces a transition of the pilot SI joint opening to an expanded post-prosthesis insertion SI joint opening.

In some embodiments, the prosthesis further comprises one of the aforementioned biologically active compositions and/or pharmacological compositions, which is disposed in one or both of the internal prosthesis engagement member lumens of the prosthesis.

In some embodiments of the invention, the prosthesis comprises an outer coating comprising one of the aforementioned biologically active agents or pharmacological agents.

In some embodiments of the invention, the outer coating comprises a poly(glycerol sebacate) (PGS) based composition.

In some embodiments, the PGS based composition comprises one or more of the aforementioned biologically active agents or pharmacological agents.

In some embodiments of the invention, the system further comprises an image capture apparatus configured and adapted to capture images reflecting positions and/or orientations of the elongated guide probe and/or defect creation assembly when disposed in the body, and, particularly, during advancement of the elongated guide probe and defect creation assembly toward and into the dysfunctional SI joint.

According to the invention, suitable image capture apparatus comprise a fluoroscope, a CT system, an ultrasound system, a radiography system, and a magnetic resonance imaging system.

In some embodiments of the invention, the system further comprises a drill guide assembly that facilitates proper placement of the elongated guide probe and pilot SI joint opening and, hence, sacrum and ilium openings thereof, and, thereby, placement of the prosthesis in the dysfunctional SI joint.

In some embodiments, the system further comprises a prosthesis extraction assembly that is adapted to remove the prosthesis from the SI joint.

BRIEF DESCRIPTION OF THE DRAWINGS

Further features and advantages will become apparent from the following and more particular description of the

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preferred embodiments of the invention, as illustrated in the accompanying drawings, and in which like referenced characters generally refer to the same parts or elements throughout the views, and in which:

FIG. 1A is a schematic illustration of a human pelvic region from an anteroposterior (AP) perspective showing the SI joints thereof;

FIG. 1B is another schematic illustration of a human pelvic region from a posterior perspective showing the adjoining sacrum and ilium bone structures, and ligamentous structures thereof;

FIG. 1C is a schematic illustration of the sacrum and coccyx from a lateral perspective showing the sacral promontory and the articular surface of sacrum;

FIG. 2A is an illustration of a SI joint from a superior perspective showing the adjoining sacrum and ilium articular surfaces;

FIG. 2B is another illustration of a SI joint from a posterior perspective showing the adjoining sacrum and ilium articular surfaces;

FIG. 2C is a further illustration of the SI joint shown in FIG. 2A showing lateral and posterior approaches to the SI joint, in accordance with the invention;

FIG. 3A is a further illustration of the SI joint shown in FIG. 2A showing an elongated guide probe of the invention positioned in the SI joint, in accordance with the invention;

FIG. 3B is a further illustration of the SI joint shown in FIG. 3A showing one embodiment of a defect creation assembly disposed proximate the SI joint, in accordance with the invention;

FIG. 3C is a perspective view of the defect creation assembly shown in FIG. 3B, in accordance with the invention;

FIG. 4A is a further illustration of the SI joint shown in FIG. 2B showing one embodiment of a pilot SI joint opening, in accordance with the invention;

FIGS. 4B and 4C are illustrations of further embodiments of SI joint openings, in accordance with the invention;

FIG. 5A is a perspective view of one embodiment of a prosthesis deployment assembly, in accordance with the invention;

FIG. 5B is a front plan view of the prosthesis deployment assembly shown in FIG. 5A, in accordance with the invention;

FIG. 5C is a left side plan view of the prosthesis deployment assembly shown in FIG. 5A, in accordance with the invention;

FIG. 5D is a top plan view of the prosthesis deployment assembly shown in FIG. 5A, in accordance with the invention;

FIG. 5E is a bottom plan view of the prosthesis deployment assembly shown in FIG. 5A, in accordance with the invention;

FIG. 5F is a front plan view of a prosthesis engagement rod of the prosthesis deployment assembly shown in FIG. 5A, in accordance with the invention;

FIG. 5G is a perspective view of the prosthesis deployment assembly shown in FIG. 5A engaged to a prosthesis of the invention, in accordance with the invention;

FIG. 6A is a perspective view of one embodiment of a prosthesis, in accordance with the invention;

FIG. 6B is a top plan view of the prosthesis shown in FIG. 6A, in accordance with the invention;

FIG. 6C is a rear plan view of the prosthesis shown in FIG. 6A, in accordance with the invention;

FIG. 6D is a front plan view of the prosthesis shown in FIG. 6A, in accordance with the invention;

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FIG. 6E is a rear perspective view of the prosthesis shown in FIG. 6A, in accordance with the invention;

FIG. 6F is a front perspective view of the prosthesis shown in FIG. 6A, in accordance with the invention;

FIG. 6G is a right-side plan view of the prosthesis shown in FIG. 6A, in accordance with the invention;

FIG. 6H is a right-side sectional plan view of the prosthesis shown in FIG. 6A, in accordance with the invention;

FIG. 6I is another rear plan view of the prosthesis shown in FIG. 6A showing the cross-sectional shape defined by the outer surface of the prosthesis, in accordance with the invention;

FIG. 7A is an illustration of the prosthesis shown in FIG. 6A inserted into the pilot SI joint opening shown in FIG. 4A and the resulting or induced post-prosthesis insertion SI joint opening, in accordance with the invention;

FIG. 7B is an illustration of the prosthesis shown in FIG. 6A inserted in the pilot SI joint opening shown in FIG. 4B and the resulting or induced post-prosthesis insertion SI joint opening, in accordance with the invention;

FIG. 8A is an illustration of the post-prosthesis insertion SI joint opening generated or induced when the prosthesis shown in FIG. 6A is inserted in the pilot SI joint opening shown in FIG. 4A, in accordance with the invention;

FIG. 8B is an illustration of the post-prosthesis insertion SI joint opening generated or induced when the prosthesis shown in FIG. 6A is inserted in the pilot SI joint opening shown in FIGS. 4B and/or 4C, in accordance with the invention;

FIG. 9A is a perspective view of one embodiment of drill guide assembly, in accordance with the invention;

FIG. 9B is a perspective view the access sleeve of the drill guide assembly shown in FIG. 9A, in accordance with the invention;

FIG. 9C is a front plan view of the access sleeve shown in FIG. 9B, in accordance with the invention;

FIG. 9D is a right-side plan view of the access sleeve shown in FIG. 9B, in accordance with the invention;

FIG. 9E is a perspective view of one embodiment of an access sleeve handle that is configured to engage the access sleeve shown in FIG. 9B, in accordance with the invention;

FIG. 9F is an end plan view of the access sleeve handle shown in FIG. 9E, in accordance with the invention;

FIG. 9G is a perspective view of the drill guide of the drill guide assembly shown in FIG. 9A, in accordance with the invention;

FIG. 9H is a front plan view of the drill guide shown in FIG. 9G, in accordance with the invention;

FIG. 9I is a top plan view of the drill guide shown in FIG. 9G, in accordance with the invention;

FIG. 9J is a bottom plan view of the drill guide shown in FIG. 9G, in accordance with the invention;

FIG. 9K is a perspective view of the guide pin of the drill guide assembly shown in FIG. 9A, in accordance with the invention;

FIG. 9L is a perspective view of the bone dislodging apparatus, i.e., drill bit, shown in FIG. 9A, in accordance with the invention;

FIG. 10A is a perspective view of one embodiment of a prosthesis extraction assembly, in accordance with the invention;

FIG. 10B is a front plan sectional view of the prosthesis extraction rod of the prosthesis extraction assembly shown in FIG. 10A, in accordance with the invention;

FIG. 10C is a partial section, front sectional plan view of the threaded end of the prosthesis extraction rod shown in FIG. 10B, in accordance with the invention;

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FIG. 10D is a perspective view of the extraction fork of the prosthesis extraction assembly shown in FIG. 10A, in accordance with the invention;

FIG. 10E is a front plan view of the extraction fork shown in FIG. 10D, in accordance with the invention;

FIG. 10F is a top plan view of the extraction fork shown in FIG. 10D, in accordance with the invention;

FIG. 10G is an exploded view of the slap hammer assembly of the prosthesis extraction assembly shown in FIG. 10A, in accordance with the invention;

FIGS. 11A and 11B are CT scan images of a patient's SI joint at six (6) months following an SI joint stabilization procedure with the prosthesis shown in FIG. 6A, in accordance with the invention; and

FIG. 12 is a graphical illustration of a patient's visual analog pain score over a six (6) week period of time following an SI joint stabilization procedure with the prosthesis shown in FIG. 6A, in accordance with the invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

Before describing the present invention in detail, it is to be understood that this invention is not limited to particularly exemplified apparatus, systems, structures or methods as such may, of course, vary. Thus, although a number of apparatus, systems and methods similar or equivalent to those described herein can be used in the practice of the present invention, the preferred apparatus, systems, structures and methods are described herein.

It is also to be understood that, although the present invention is described and illustrated in connection with sacroiliac (SI) joint stabilization, fixation and fusion procedures, the invention is not limited to such procedures. According to the invention, the apparatus, systems and methods of the invention can also be employed to stabilize and/or fuse other articulating bone structures, including, without limitation, spinal vertebrae, tarsal bones and the like.

It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments of the invention only and is not intended to be limiting.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one having ordinary skill in the art to which the invention pertains.

Further, all publications, patents and patent applications cited herein, whether supra or infra, are hereby incorporated by reference in their entirety.

As used in this specification and the appended claims, the singular forms "a," "an" and "the" include plural referents unless the content clearly dictates otherwise. Thus, for example, reference to "an incision" includes two or more incisions and the like.

Further, ranges can be expressed herein as from "about" or "approximately" one particular value, and/or to "about" or "approximately" another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent "about" or "approximately", it will be understood that the particular value forms another embodiment. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

It is also understood that there are a number of values disclosed herein, and that each value is also herein disclosed

as “about” or “approximately” that particular value in addition to the value itself. For example, if the value “10” is disclosed, then “approximately 10” is also disclosed. It is also understood that when a value is disclosed that “less than or equal to” the value, “greater than or equal to the value” and possible ranges between values are also disclosed, as appropriately understood by the skilled artisan. For example, if the value “10” is disclosed then “less than or equal to 10” as well as “greater than or equal to 10” is also disclosed.

Definitions

The terms “sacroiliac joint”, “SI joint”, “sacroiliac junction” and “SI junction” are used interchangeably herein, and mean and include any region proximate to articulating regions of the sacrum and ilium bone structures and, hence, a junction between and defined by sacrum and ilium bone structures.

The term “dysfunctional” as used in connection with a SI joint, means and includes a physiological abnormality, disorder or impairment of an SI joint, including, but limited to, traumatic fracture dislocation of the pelvis, degenerative arthritis, sacroiliitis, i.e., an inflammation or degenerative condition of the SI joint; osteitis condensans ilii, and other degenerative conditions of SI joint bone structures.

The terms “articular surface” and “articulating surface” are used interchangeably herein in connection with bone structures; particularly, the sacrum and ilium bone structures, and mean and include a surface of a bone structure that forms an articulating junction (i.e., a synovial joint) with an adjacent bone structure, e.g., the articular surfaces of the sacrum and ilium bone structures.

The terms “fusion” and “arthrodesis” are used interchangeably herein in connection with bone structures, and mean and include partial or complete immobilization of adjacent bone structures; particularly, the sacrum and ilium bone structures.

The term “stabilization”, as used herein, means and includes reinforcing, e.g., supporting, or modulating motion of adjacent articular bone structures; particularly, the sacrum and ilium bone structures. The term “stabilization”, thus, in some instances, means and includes fusion and arthrodesis of adjacent bone structures.

The term “prosthesis”, as used herein in connection with bone structures, means and includes a system or apparatus configured and adapted to stabilize or modulate motion of articulating bone structures; particularly, the sacrum and ilium bone structures.

The term “biodegradable”, as used herein, means the ability of a material; particularly, a polymer or adhesive, to breakdown and be absorbed within the physiological environment of a SI joint and/or a structure associated therewith, including sacrum and ilium bone structures, by one or more physical, chemical, or cellular processes.

Biodegradable polymers, according to the invention, thus include, without limitation, polylactide polymers (PLA), copolymers of lactic and glycolic acids, including poly(lactic-co-glycolic) acid (PLGA) and poly(ϵ -caprolactone-co-L-lactic) acid (PCL-LA); glycine/PLA co-polymers, polyethylene oxide (PEO)/PLA block copolymers, acetylated polyvinyl alcohol (PVA)/polycaprolactone copolymers, poly(glycerol sebacate) (PGS) and its derivatives, including poly(glycerol-co-sebacate acrylate) (PGSA); poly (polyol sebacate) (PPS), poly(xylitol sebacate) (PXS), poly (xylitol glutamate sebacate) (PXGS), hydroxybutyrate-hydroxyvalerate copolymers, polyesters such as, but not limited to, aspartic acid and different aliphatic diols; poly

(alkylene tartrates) and their copolymers with polyurethanes, polyglutamates with various ester contents and with chemically or enzymatically degradable bonds, other biodegradable nonpeptide polyamides, amino acid polymers, polyanhydride drug carriers such as, but not limited to, poly(sebacic acid) (PSA); aliphatic-aromatic homopolymers, and poly(anhydride-co-imides), poly(phosphoesters) by matrix or pendant delivery systems, poly(phosphazenes), poly(iminocarbonate), crosslinked poly(ortho ester), hydroxylated polyester-urethanes, or the like.

Biodegradable adhesives, according to the invention, thus include, without limitation, poly(glycerol-co-sebacate acrylate) (PGSA), poly(L-glutamic acid)-based compositions, poly(γ -glutamic acid)-based compositions, poly(alkyl cyanoacrylate)-based compositions, polyacrylic acid-based compositions, including polyacrylic acid crosslinked with pentaerythritol and/or allyl sucrose, polyacrylic acid crosslinked with divinyl glycol, and combinations thereof; fibrin-based compositions, collagen-based compositions, including collagen/poly(L-glutamic acid) compositions; albumin-based compositions, including BioGlue® (comprises purified bovine serum albumin (BSA) and glutaraldehyde); cyanoacrylate compositions, including butyl-2-cyanoacrylate adhesives (e.g., Indermil®, Histoacryl®, Histoacryl® Blue, and LiquiBand®) and octyl-2-cyanoacrylate adhesives (e.g., Dermabond®, SurgiSeal™, LiquiBand® Flex, and Octyl-Seal); poly(ethylene glycol) (PEG) based compositions, including FocalSeal®, Progel™, Duraseal™, DuraSeal™ Xact, Coseal® and ReSure Sealant; polysaccharide-based compositions, polypeptide-based compositions, and combinations thereof.

The term “osteogenic composition”, as used herein, means and includes an agent or composition that induces or modulates an osteogenic physiological or biological process, or cellular activity, e.g., induces proliferation, and/or growth and/or remodeling and/or regeneration of bone or osseous tissue.

The term “osteogenic composition” thus means and includes, without limitation, the following osteogenic materials and compositions comprising same: demineralized bone matrix, autograft bone material, allograft bone material, xenograft bone material, polymethyl-methacrylate, calcium-based bone void filler material, including hydroxyapatite (HA) and tricalcium phosphate; and combinations or mixtures thereof.

The term “osteogenic composition” also means and includes, without limitation, the following polymer materials and compositions comprising same: poly(glycerol sebacate) (PGS), poly(glycerol-co-sebacate) acrylate (PGSA) and co-polymers, such as poly(glycerol sebacate)-co-poly(ethylene glycol) (PGS-PEG); and/or composites thereof; e.g., PGS-hydroxyapatite (HA) composites and PGS-poly(ϵ -caprolactone) (PGS-PCL) composites.

The term “osteogenic composition” also means and includes, without limitation, acellular extracellular matrix (ECM) derived from mammalian tissue sources.

The term “osteogenic composition” thus means and includes, without limitation, acellular ECM derived from bone or osseous tissue, small intestine submucosa (SIS), epithelium of mesodermal origin, i.e., mesothelial tissue, placental tissue, omentum tissue, and combinations thereof.

The terms “biologically active agent” and “biologically active composition” are used interchangeably herein, and mean and include agent or composition that induces or modulates a physiological or biological process, or cellular activity, e.g., induces proliferation, and/or growth and/or regeneration of tissue, including osseous tissue.

The terms “biologically active agent” and “biologically active composition”, as used herein, thus include agents and compositions that can be varied in kind or amount to provide a therapeutic level effective to mediate the formation or healing of osseous tissue, cartilage and connective tissue, e.g., tendons and ligaments. The term “biologically active composition”, in some instances, thus means and includes an “osteogenic composition.”

The terms “biologically active agent” and “biologically active composition” thus mean and include, without limitation, the following bone morphogenic proteins (BMPs) and compositions comprising same: BMP-1, BMP2a, BMP2b, BMP3, BMP4, BMP5, BMP6, BMP7 (also referred to as osteogenic protein 1 (OP-1)) and BMP8a.

The terms “biologically active agent” and “biologically active composition” also mean and include, without limitation, the following biological agents and compositions comprising same: platelet derived growth factor (PDGF), an insulin-like growth factor (IGF), including IGF-1 and IGF-2; basic fibroblast growth factor (bFGF) (also referred to as FGF2), transforming growth factor- β (TGF- β), including, TGF- β 1 and TGF- β 2, a growth hormone (GH), parathyroid hormone (PTH, including PTH1-34), transforming growth factor- α (TGF- α), granulocyte/macrophage colony stimulating factor (GM-CSF), epidermal growth factor (EGF), growth and differentiation factor-5 (GDF-5), vascular endothelial growth factor (VEGF), angiogenin, angiopoietin-1, del-1, follistatin, granulocyte colony-stimulating factor (G-CSF), hepatocyte growth factor/scatter factor (HGF/SF), interleukin-8 (IL-8), interleukin-10 (IL-10), leptin, midkine, placental growth factor, platelet-derived endothelial cell growth factor (PD-ECGF), platelet-derived growth factor-BB (PDGF-BB), pleiotrophin (PTN), progranulin, proliferin, a matrix metalloproteinase (MMP), angiopoietin 1 (ang1), angiopoietin 2 (ang2) and delta-like ligand 4 (DLL4).

The terms “biologically active agent” and “biologically active composition” also mean and include, without limitation, the following cells and compositions comprising same: bone marrow-derived progenitor cells, bone marrow stromal cells (BMSCs), osteoprogenitor cells, osteoblasts, osteocytes, osteoclasts, committed or partially committed cells from the osteogenic or chondrogenic lineage, hematopoietic stem cells, chondrocytes, chondrogenic progenitor cells (CPCs), mesenchymal stem cells (MSCs) and embryonic stem cells.

The terms “biologically active agent” and “biologically active composition” also mean and include an “extracellular vesicle (EV)”, “exosome”, “microsome” or “micro-vesicle”, which are used interchangeably herein, and mean and include a biological structure formed from a hydrocarbon monolayer or bilayer configured to contain or encase a composition of matter.

The terms “extracellular vesicle (EV)”, “exosome”, “microsome” and “micro-vesicle” thus include, without limitation, a biological structure formed from a lipid layer configured to contain or encase biologically active agents and/or combinations thereof.

The terms “extracellular vesicle (EV)”, “exosome”, “microsome” and “micro-vesicle” also include, without limitation, EVs derived from the aforementioned cells and compositions comprising same, e.g., BMSC-derived EVs.

The terms “pharmacological agent” and “active agent” are used interchangeably herein, and mean and include an agent, drug, compound, composition or mixture thereof, including its formulation, which provides some therapeutic, often beneficial, effect. This includes any physiologically or phar-

macologically active substance (or composition comprising same) that produces a localized or systemic effect or effects in animals, including warm blooded mammals.

The terms “pharmacological agent” and “active agent” thus mean and include, without limitation, the following osteoinductive agents and compositions comprising same: icaritin, tumor necrosis factor alpha (TNF- α) inhibitors, including etanercept and infliximab, disease-modifying anti-rheumatic drugs (DMARDs), including methotrexate and hydroxychloroquine, antibiotics, anti-viral agents, steroidal anti-inflammatories, non-steroidal anti-inflammatories, anti-thrombotic agents, including anti-coagulants and anti-platelet agents, and vasodilating agents.

The terms “pharmacological agent” and “active agent” further mean and include, without limitation, the following bisphosphonate agents and compositions comprising same: risedronate (Actonel®), alendronate (Fosamax®), ibandronate (Boniva®), zoledronic acid (Reclast®), pamidronate (Aredia®) and etidronate (Didronel®).

The terms “pharmacological agent” and “active agent” further mean and include, without limitation, the following antibiotics and compositions comprising same: penicillin, carboxypenicillins, such as ticarcillin; tetracyclines, such as minocycline; gentamicin, vancomycin, ciprofloxacin, amikacin, aminoglycosides, cephalosporins, clindamycin, erythromycin, fluoroquinolones, macrolides, azolides, metronidazole, trimethoprim-sulfamethoxazole, polymyxin B, oxytetracycline, tobramycin, cefazolin and rifampin.

The terms “anti-inflammatory” and “anti-inflammatory agent” are also used interchangeably herein, and mean and include a “pharmacological agent”, which, when a therapeutically effective amount is administered to a subject, prevents or treats bodily tissue inflammation, i.e., the protective tissue response to injury or destruction of tissues, which serves to destroy, dilute, or wall off both the injurious agent and the injured tissues.

Anti-inflammatory agents thus include, without limitation, dexamethasone, betamethasone, prednisone, prednisolone, methylprednisolone sodium succinate, methylprednisolone, cortisone, ketorolac, diclofenac and ibuprofen.

The terms “pharmacological agent” and “active agent” further mean and include, without limitation, the following metal-based antimicrobials and compositions comprising same: silver particles, copper particles, cobalt particles, nickel particles, zinc particles, zirconium particles, molybdenum particles, lead particles and mixtures thereof.

As indicated above, the term “pharmacological composition”, as used herein, means and includes a composition comprising a “pharmacological agent” and “active agent”.

The term “therapeutically effective”, as used herein, means that the amount of the “pharmacological agent” and/or “pharmacological composition” and/or “biologically active agent” and/or “biologically active composition” administered is of sufficient quantity to ameliorate one or more causes, symptoms, or sequelae of a disease or disorder. Such amelioration only requires a reduction or alteration, not necessarily elimination, of the cause, symptom, or sequelae of a disease or disorder.

The terms “patient” and “subject” are used interchangeably herein, and mean and include warm blooded mammals, humans and primates; avians; domestic household or farm animals, such as cats, dogs, sheep, goats, cattle, horses and pigs; laboratory animals, such as mice, rats and guinea pigs; fish; reptiles; zoo and wild animals; and the like.

The terms “one embodiment”, “one aspect”, and “an embodiment” and “an aspect”, as used herein, means that a particular feature, structure, or characteristic described in

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connection with the embodiment may be included in at least one embodiment and not that any particular embodiment is required to have a particular feature, structure or characteristic described herein unless set forth in the claim.

The phrase “in one embodiment” or similar phrases employed herein do not limit the inclusion of a particular element of the invention to a single embodiment. The element may thus be included in other, or all embodiments discussed herein.

The term “substantially”, as used herein, means and includes the complete or nearly complete extent or degree of an action, characteristic, property, state, structure, item, or result to function as indicated. For example, an object that is “substantially” enclosed would mean that the object is either completely enclosed or nearly completely enclosed. The exact allowable degree of deviation from absolute completeness may in some cases depend on the specific context, such that enclosing nearly all the length of a lumen would be substantially enclosed, even if the distal end of the structure enclosing the lumen had a slit or channel formed along a portion thereof.

Use of the term “substantially” is equally applicable when used in a negative connotation to refer to the complete or near complete lack of an action, characteristic, property, state, structure, item, or result. For example, structure which is “substantially free of” a bottom would either completely lack a bottom or so nearly completely lack a bottom that the effect would be effectively the same as if it completely lacked a bottom.

The term “comprise” and variations of the term, such as “comprising” and “comprises,” means “including, but not limited to” and is not intended to exclude, for example, other components, elements or steps.

The following disclosure is provided to further explain in an enabling fashion the best modes of performing one or more embodiments of the present invention. The disclosure is further offered to enhance the understanding and appreciation for the inventive principles and advantages thereof, rather than to limit in any manner the invention. The invention is defined solely by the appended claims, including any amendments made during the pendency of this application, and all equivalents of those claims as issued.

As indicated above, the present invention is directed to minimally-invasive methods, systems and apparatus for stabilizing dysfunctional SI joints.

In some embodiments of the invention, there are thus provided minimally-invasive systems for stabilizing dysfunctional SI joints. As indicated above, in a preferred embodiment, the minimally-invasive systems (also referred to herein as “minimally-invasive SI joint stabilization systems”) can be readily employed in minimally-invasive methods or procedures to stabilize dysfunctional SI joints via a posterior approach.

As indicated above, SI joint stabilization, including minimally-invasive SI joint stabilization, typically comprises surgical placement of a prosthesis proximate to or in a dysfunctional SI joint via anterior, lateral and posterior approaches to the SI joint.

From the perspective of FIG. 1A, an anterior approach to the SI joint 6 (and, hence, a dysfunctional SI joint) would be substantially perpendicular to the page upon which FIG. 1A is printed.

Referring now to FIG. 2A there is shown a close-up illustration of a portion of the leftmost SI joint 6 illustrated in FIG. 1A. For illustrative simplicity, a uniform layer of cortical bone 8 is shown adjacent a deeper layer of trabecular

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bone 10 on both of the depicted sacrum 2 and ilium 4 portions. However, in actuality, such layers are far less uniform and homogeneous.

Referring now to FIG. 2B, there is shown a view of the same structure from a different posterior perspective. From the perspective of FIG. 2B, a posterior approach to the SI joint 6 (and, hence, a dysfunctional SI joint) would be substantially perpendicular to the page upon which FIG. 2B is printed. Indeed, referring to FIG. 2C, a variation similar to that depicted in FIG. 2A is illustrated, showing an approximate approach vector for a lateral approach to the SI joint 6 versus a posterior approach, using the orientation paradigms introduced in FIGS. 1A and 2A-2C. Such paradigm is used to illustrate various embodiments of the subject invention in various figures that follow FIGS. 1A and 2A-2C.

As indicated above, a major disadvantage associated with many conventional anterior or lateral approaches to a dysfunctional SI joint is that muscles and ligaments are typically disrupted and often damaged. Nerves and blood vessels are also susceptible to damage during such SI joint stabilization methods.

In contrast, a posterior approach to a dysfunctional SI joint is much less invasive. Indeed, less tissue and fewer muscles are disrupted, and nerves and large blood vessels are avoided.

As indicated above, in a preferred embodiment of the invention, the system for stabilizing a dysfunctional SI joint comprises a tool assembly and a prosthesis.

Referring now to FIGS. 3A-3C, 4A-4C and 5A-5B, a preferred tool assembly of the invention will be described in detail. As illustrated in FIGS. 3A-3C and 5A-5B, in a preferred embodiment, the tool assembly comprises an elongated guide probe 20, an SI joint opening or defect creation assembly (referred to hereinafter as “defect creation assembly”) 30, and prosthesis deployment assembly 50.

Referring first to FIGS. 3A and 3B, there is shown a preferred embodiment of an elongated guide probe 20 of the invention. As illustrated in FIGS. 3A and 3B, the elongated guide probe 20 comprises proximal and distal ends 22, 24.

As further illustrated in FIG. 3B, and set forth in Co-pending priority U.S. application Ser. No. 13/857,977 and Co-pending U.S. application Ser. No. 17/463,779, filed on Sep. 1, 2021, which are expressly incorporated by reference herein, the elongated guide probe 20 is sized and configured to be positioned in the dysfunctional SI joint and function as a guide for advancing the defect creation assemblies of the invention; particularly, defect creation assembly 30, into dysfunctional SI joints and placement of a prosthesis therein.

Referring now to FIGS. 3B and 3C, there is shown a preferred embodiment of the defect creation assembly 30 (referred to as an “osteotome/cannulation assembly”) in Co-pending priority U.S. application Ser. No. 13/857,977).

As also set forth in Co-pending U.S. application Ser. Nos. 13/857,977 and 17/463,779, the defect creation assembly 30 is configured and adapted to create pre-determined, surgically produced open spaces or defects in the dysfunctional SI joint (referred to herein after as “pilot SI joint openings”) to accommodate placement of a prosthesis of the invention therein.

As illustrated in FIGS. 3B and 3C, the defect creation assembly 30 comprises a housing 32, having a longitudinal axis LA₁, a proximal end 34, a distal end 36, and a guide member lumen 38 that extends through the defect creation assembly 30.

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As further illustrated in FIG. 3B, the guide member lumen 38 adapted to receive said guide probe 20 therein, whereby the defect creation assembly 30 is allowed to slidably translate or be advanced along the guide probe 20 to position the defect creation assembly 30 proximate to a dysfunctional SI joint site.

In a preferred embodiment of the invention, the defect creation assembly 30 further comprises a bone dislodging apparatus or system 40 disposed on the defect creation assembly distal end 36, which is configured and adapted to dislodge portions of osseous tissue, i.e., bone, proximate and in the dysfunctional SI joint.

As set forth in Co-pending U.S. application Ser. No. 17/463,779, the bone dislodging system 40 can comprise various bone dislodging apparatus, such as a drill assembly and associated drill bit or orthopedic burr, which can be operated manually, pneumatically, or electromechanically. In a preferred embodiment, the bone dislodging system 40 comprises a drill assembly and associated drill bit.

As shown in greater detail in FIG. 3C, according to the invention, the distal end 36 of the defect creation assembly 30 may also comprise one or more teeth or apices 42 configured to assist with creation of a pilot SI joint opening in SI joint bone structures, i.e., sacrum or ilium bone structures.

As indicated above, in a preferred embodiment, the defect creation assemblies of the invention (and, particularly, defect creation assembly 30) are configured and adapted to create pilot SI joint openings in SI joint bone structures to accommodate placement of a prosthesis of the invention therein.

It is however, to be understood that defect creation assembly 30 described herein, is but one embodiment of a defect creation assembly that can be employed within the scope of the invention to create pilot SI joint openings in SI joint bone structures. Indeed, as indicated above and discussed in detail below, various conventional apparatus and systems, such as a surgical drill, can also be employed within the scope of the invention to create pilot SI joint openings of the invention in SI joint bone structures.

As also set forth in Co-pending U.S. application Ser. Nos. 13/857,977 and 17/463,779, the defect creation assemblies of the invention, including defect creation assembly 30, are configured and adapted to create pilot SI joint openings in SI joint bone structures of various sizes and configurations. Illustrative are the pilot SI joint openings depicted in FIGS. 11C-11E of Co-pending priority U.S. application Ser. No. 13/857,977.

Referring now to FIG. 4A, there is shown one embodiment of a preferred pilot SI joint opening of the invention (denoted "100") that can be created with the defect creation assemblies of the invention; particularly, defect creation assembly 30.

As illustrated in FIG. 4A, the pilot SI joint opening 100 comprises a three-dimensional opening comprising first and second lobe regions 103, 104; the first lobe region 103 being disposed in the sacrum 2 and comprising a sacrum opening three-dimensional shape, and the second lobe region 104 being disposed in the ilium 4 and comprising an ilium opening three-dimensional shape.

As further illustrated in FIG. 4A, the three-dimensional pilot SI joint opening 100 preferably comprises an SI joint opening cross-sectional shape in a plane that intersects the sacrum 2 and ilium 4 bone structures; the plane being substantially perpendicular to the longitudinal axis LA_1 of the defect creation assembly 30 when the defect creation assembly 30 is disposed in a defect creation position in the

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dysfunctional SI joint. The three-dimensional pilot SI joint opening cross-sectional shape thus comprises the sacrum opening three-dimensional shape and ilium opening three-dimensional shape.

In some embodiments, the three-dimensional pilot SI joint opening 100 is defined in part by at least one noncircular cross-sectional shaped region (denoted "105") in the noted plane.

As additionally illustrated in FIG. 4A, the three-dimensional pilot SI joint opening 100, i.e., cross-sectional shape thereof, also defines a cross-sectional area of the three-dimensional pilot SI joint opening cross-sectional shape (denoted " A^2_{I-1} ")

The three-dimensional pilot SI joint opening 100, i.e., cross-sectional shape thereof, also comprises a longitudinal axis (denoted " LA_2 ") in the plane that intersects the sacrum 2 and ilium 4 and an initial pilot SI joint opening length along the axis LA_2 .

Referring now to FIG. 4B, there is shown a further pilot SI joint opening of the invention (denoted "200") that can be created with the defect creation assemblies of the invention; particularly, defect creation assembly 30.

As illustrated in FIG. 4B, the pilot SI joint opening 200 comprises two three-dimensional pilot or guide portions or regions 203, 204; the first guide portion 203 being disposed in the sacrum 2 and the second guide portion 204 being disposed in the ilium 4.

According to the invention, the sacrum and ilium guide portions 203, 204 can comprise various configurations, e.g., cross-sectional shapes, and sizes to, as discussed in detail below, accommodate insertion of defined regions of a prosthesis of the invention therein and transition of the sacrum and ilium guide portions 203, 204 from pilot or first configurations and sizes to expanded second configurations and sizes when the prosthesis is inserted therein.

As set forth in Co-pending U.S. application Ser. No. 17/463,779, the sacrum and ilium guide portions 203, 204 can also be disposed at various locations in the sacrum 2 and ilium 4, such as shown in FIGS. 4A and 4B.

As illustrated in FIG. 4B, in a preferred embodiment, the sacrum and ilium guide portions 203, 204 of the pilot SI joint opening 200 created by the defect creation assembly 30 of the invention comprise substantially circular cross-sectional shapes.

As further illustrated in FIG. 4B, the sacrum and ilium guide portions 203, 204 of the pilot SI joint opening 200, i.e., cross-sectional shape thereof, define cross-sectional areas of the sacrum and ilium guide portions 203, 204 (denoted " A^2_{I-2} " and " A^2_{I-3} ", respectively).

In a preferred embodiment, the sacrum and ilium guide portions 203, 204 of the pilot SI joint opening 200 are disposed on a plane that similarly intersects the sacrum 2 and ilium 4.

Referring now to FIGS. 5A-5G, there is shown a preferred embodiment of a prosthesis deployment assembly of the invention (denoted "50" in FIG. 5G).

As also set forth in Co-pending U.S. application Ser. No. 17/463,779, in a preferred embodiment, the prosthesis deployment assembly 50 comprises prosthesis engagement means configured and adapted to connect the prosthesis deployment assembly 50 to prostheses of the invention (prosthesis 70 shown in FIG. 5G) and guide the prostheses into pilot SI joint openings created by the defect creation assembly 30.

As illustrated in FIGS. 5A-5C, the prosthesis deployment assembly 50 comprises an elongated guide member 51 comprising proximal and distal ends 52, 54.

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As further illustrated in FIGS. 5B and 5E, the elongated guide member 51 further comprises a prosthesis guide pin 56 that extends from the guide member distal end 54. As discussed in detail below and shown in FIG. 5G, the prosthesis guide pin 56 is sized and configured to seat in an internal prosthesis engagement member lumen 86a or 86b of the preferred prosthesis 70 of the invention.

As illustrated in FIGS. 5A, 5D and 5E, the elongated guide member 51 further comprises an internal lumen 58 that extends from the proximal end 52 of the elongated guide member 51 to the distal end 54 of the elongated guide member 51.

As illustrated in FIG. 5G, in a preferred embodiment of the invention, the internal lumen 58 is sized and configured to receive the prosthesis engagement rod 60 (i.e., prosthesis engagement means) of the prosthesis deployment assembly 50, discussed below.

Referring now to FIG. 5F, there is shown a preferred embodiment of a prosthesis engagement rod 60 of the invention. As illustrated in FIG. 5F, the prosthesis engagement rod 60 comprises a proximal end 62 and a threaded distal end 64, which, as discussed in detail below, is sized and configured to threadably engage an internal prosthesis engagement member lumen of a prosthesis of the invention, e.g., internal prosthesis engagement member lumens 86a and/or 86b of prosthesis 70.

As further illustrated in FIG. 5F, in a preferred embodiment, the proximal end 62 of the prosthesis engagement rod 60 comprises a knurled configuration to facilitate threading the prosthesis engagement rod 60 into an internal prosthesis engagement member lumen of a prosthesis of the invention.

Referring back to FIGS. 5A and 5B, to further facilitate threading the prosthesis engagement rod 60 into an internal prosthesis engagement member lumen of a prosthesis of the invention, in a preferred embodiment, the elongated guide member 51 further comprises an access port 57 that provides access to the knurled proximal end 62 of the prosthesis engagement rod 60 when positioned in the internal lumen 58 of the elongated guide member 51, as shown in FIG. 5G.

According to the invention, the system for stabilizing dysfunctional SI joints can comprise various prostheses, which are configured and adapted to be inserted into pilot SI joint openings created by a defect creation assembly of the invention.

Suitable prostheses that are configured and adapted to be inserted into a pilot SI joint opening created by a defect creation assembly of the invention are set forth in Co-pending priority application Ser. Nos. 13/857,977 and 17/463,779, which is expressly incorporated by reference herein.

According to the invention, the prostheses illustrated in FIGS. 12A-12C, 13A-13B, 14A-14C and 15A-15D of Co-pending priority application Ser. No. 13/857,977 are suitable for insertion into pilot SI joint openings of the invention (i.e., SI joint openings 100, 200 described above) in a SI joint, and into and through articular cartilage and cortical bone (and trabecular bone), which define the SI joint.

Referring now to FIGS. 6A-6I, there is shown a preferred prosthesis 70 of the invention, which is particularly suitable for placement in pilot SI joint openings of the invention (particularly, pilot SI joint openings 100 and 200) in a SI joint, and into and through articular cartilage and bone structures (i.e., cortical and trabecular bone), which define the SI joint.

As illustrated in FIGS. 6A, 6E and 6F, the prosthesis 70 comprises a biocompatible and, hence, implantable member comprising proximal and distal ends 72, 74, and first and

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second elongated partially cylindrical sections 76a, 76b connected to a bridge section 78, whereby the prosthesis 70 comprises a continuous exterior surface comprising first and second partially cylindrical surface regions 77a, 77b.

As further illustrated in FIGS. 6A, 6E and 6F, the first and second partially cylindrical sections 76a, 76b comprise proximal and distal ends 79a, 79b. The bridge section 78 similarly comprises proximal and distal ends 81a, 81b.

According to the invention, the prosthesis 70 can comprise any suitable length from the proximal ends 79a to the distal ends 79b of the partially cylindrical sections 76a, 76b. In some embodiments, the prosthesis 70 comprises a length in the range of 20-50 mm, more preferably, a length in the range of 30-40 mm.

As illustrated in FIGS. 6C, 6E, 6F and FIGS. 4A and 4B, the first partially cylindrical surface region 77a preferably comprises a partially cylindrical surface region shape that corresponds to at least a portion of the first lobe region 103 of the pilot SI joint opening 100 and/or the sacrum guide portion 203 of the pilot SI joint opening 200, and/or the second lobe region 104 of the pilot SI joint opening 100 and/or the ilium guide portion 204 of the pilot SI joint opening 200, depending on the entry position of the prosthesis 70 into the pilot SI joint openings 100, 200.

The second partially cylindrical surface region 77b similarly preferably comprises a partially cylindrical surface region shape that corresponds to at least a portion of the first lobe region 103 of the pilot SI joint opening 100 and/or the sacrum guide portion 203 of the pilot SI joint opening 200, or the second lobe region 104 of the pilot SI joint opening 100 and/or the ilium guide portion 204 of the pilot SI joint opening 200, again depending on the entry position of the prosthesis 70 into the pilot SI joint openings 100, 200.

As illustrated in FIGS. 6A, 6B, 6F-6H, the distal end 81b of the bridge section 78 preferably comprises a taper region 82, which is configured and adapted to disrupt, i.e., cut into and through, articular cartilage and cortical bone 8 (and, in some aspects, trabecular bone 10), which define a SI joint.

According to the invention, the taper region 82 of the bridge section 78 can comprise various configurations including, without limitation, X-bevel, wedge-shaped or bevel, including top and bottom wedge bevels, Y-bevel, including top and bottom Y-bevels, and K-bevel configurations.

In some embodiments of the invention, the taper region 82 comprises two angled regions that intersect at a central point 83, i.e., pointed proximate the mid-region of the bridge section 78, such as shown in FIGS. 6A and 6F. In some embodiments, the taper region 82 comprises a single angled or sloped region defining a plane that intersects the plane defined by the bottom surface of the prosthesis 70, i.e., wedge shaped or bevel configuration.

In a preferred embodiment, the distal ends 79b of the first and second elongated partially cylindrical sections 76a, 76b also comprise tapered regions 84a, 84b, which facilitate (i) insertion of the distal ends 79b of the first and second elongated partially cylindrical sections 76a, 76b into the first and second lobe regions 103, 104 of the pilot SI joint opening 100 and/or the sacrum and ilium guide portions 203, 204 of the pilot SI joint opening 200, and (ii) as discussed in detail below, in some embodiments, transition of the pilot SI joint opening 100 from a first configuration and size (and, hence, cross-sectional area, i.e., A^2_1 -1 shown in FIG. 4A) to a second expanded configuration and size (and, hence, cross-sectional area, i.e., A^2_2 -4 shown in FIG. 8A) when the prosthesis 70 is inserted therein, and transition of the sacrum and ilium guide portions 203, 204 of pilot SI opening 200

from first configurations and sizes (and, hence, cross-sectional areas, i.e., A^2_i-2 and A^2_i-3 shown in FIG. 4B) to expanded second configurations and sizes (and, hence, cross-sectional areas, i.e., A^2-5 and A^2-6 shown in FIG. 8B) when the prosthesis **70** is inserted therein.

As illustrated in FIGS. 6C, 6E, and 6H, the first elongated partially cylindrical section **76a** of the prosthesis **70** comprises an internal prosthesis engagement member lumen **86a** that extends from the proximal end **79a** of the first elongated partially cylindrical section **76a**.

As illustrated in FIGS. 6C and 6E, the second elongated partially cylindrical section **76b** of the prosthesis **70** also comprises an internal prosthesis engagement member lumen **86b** that extends from the proximal end **79a** of the first elongated partially cylindrical section **76b**.

In a preferred embodiment, the internal prosthesis engagement member lumens **86a**, **86b** of the prosthesis **70** are sized and configured to receive the prosthesis guide pin **56** of the prosthesis deployment assembly **50** and, as discussed below, the prosthesis engagement rod **60** of the prosthesis deployment assembly **50**.

As set forth in Co-pending U.S. application Ser. No. 17/463,779 and illustrated in FIGS. 6E and 6G, in a preferred embodiment, the internal prosthesis engagement member lumens **86a**, **86b** of the first and second elongated partially cylindrical sections **76a**, **76b** comprise a threaded region **87** proximate the proximal end **79a** that is sized and configured to receive and threadably engage the threaded distal end **64** of the prosthesis engagement rod **60** of the prosthesis deployment assembly **50** and, as discussed in detail below, the prosthesis extraction rods or screws **602a**, **602b** of the prosthesis extraction assembly **600**.

In a preferred embodiment, the internal prosthesis engagement lumens **86a**, **86b** are also configured to receive agents and compositions that further facilitate adhesion of the prosthesis **70** to the pilot SI openings **100**, **200** of the invention and, thereby, sacrum and/or ilium bone structures, and the aforementioned biologically active agents and compositions, including osteogenic agents and compositions, and pharmacological agents and compositions that facilitate osseous or bone tissue ingrowth into the prosthesis **70** and healing of the SI joint bone structures.

Referring back to FIGS. 6A and 6B, in a preferred embodiment, the prosthesis **70** further comprises a plurality of slots **90** and holes **92**, which preferably are in communication with the internal prosthesis engagement member lumens **86a**, **86b**.

In a preferred embodiment, the agents and compositions referenced above are adapted to extrude through the slots **90** and holes **92** of the prosthesis **70** when the prosthesis **70** is inserted in a pilot SI joint opening (i.e., pilot SI joint openings **100** or **200**), to, as indicated above, (i) further facilitate adhesion of the prosthesis **70** to the pilot SI openings **100**, **200** of the invention and, thereby, sacrum and/or ilium, and (ii) facilitate osseous or bone tissue ingrowth into the prosthesis **70** and healing of the SI joint bone structures.

Referring now to FIG. 6I, according to the invention, the continuous exterior surface of the prosthesis **70**, which is illustrated in FIGS. 6C and 6D, defines a prosthesis cross-sectional shape (denoted " P_{CSS} ") having a longitudinal axis LA_3 .

According to one embodiment of the invention, the length of the prosthesis cross-sectional shape P_{CSS} along longitudinal axis LA_3 is greater than the length of the pilot SI joint opening **100**, i.e., cross-sectional shape thereof illustrated in FIG. 4A, along the longitudinal axis LA_2 thereof, whereby,

when the prosthesis **70** is inserted into pilot SI joint opening **100**, as illustrated in FIG. 7A, the pilot SI opening **100** transitions to a post-prosthesis insertion SI joint opening **300** comprising a larger cross-sectional length shape that corresponds to the length of the prosthesis cross-sectional shape P_{CSS} .

As illustrated in FIG. 8A, in a preferred embodiment, when the prosthesis **70** is inserted into pilot SI joint opening **100**, the cross-sectional area of the post-prosthesis insertion SI joint opening **300** also comprises a cross-sectional area (denoted " A^2-4 ") that is greater than the cross-sectional area A^2_i-1 of the pilot SI joint opening **100**.

As further illustrated in FIG. 8A, the noncircular region **105** of pilot SI joint opening **100** also transitions to a much larger noncircular region (denoted "**305**"), which is achieved by virtue of the tapered bridge section **78** of the prosthesis **70** cutting into and through the articular cartilage and cortical bone **8**, which define the SI joint **6**, and the trabecular bone **10** proximate the SI joint **6**.

According to the embodiment of the invention, when the prosthesis **70** is inserted into pilot SI joint opening **200**, as illustrated in FIG. 7B, the pilot SI joint opening **200** similarly transitions to a post-prosthesis insertion SI joint opening **400**, wherein, as illustrated in FIG. 8B, the cross-sectional areas of the post-prosthesis sacrum and ilium guide portions of the post-prosthesis insertion SI joint opening **400** (now denoted "**402**" and "**404**", respectively) comprise greater cross-sectional areas (denoted " A^2-5 " and " A^2-6 ").

As further illustrated in FIG. 8B, the post-prosthesis insertion SI joint opening **400** also comprises a noncircular region (denoted "**405**"), which is similarly achieved by virtue of the tapered bridge section **78** of the prosthesis **70** cutting into and through the articular cartilage and cortical bone **8**, which define the SI joint **6**, and the trabecular bone **10** proximate the SI joint **6**.

As illustrated in FIGS. 6I, 8A and 8B, the post-prosthesis insertion SI joint openings **300**, **400** also comprise cross-sectional shapes that correspond to the prosthesis cross-sectional shape " P_{CSS} " defined by the outer surface of the prosthesis **70**, including the first and second elongated partially cylindrical sections **76a**, **76b** and bridge section **78**.

In a preferred embodiment of the invention, to achieve sufficient expansion of the pilot SI joint openings **100**, **200** when the prosthesis **70** is inserted therein, preferably, the cross-sectional areas of the regions defined by the first and second elongated partially cylindrical sections **76a**, **76b** of the prosthesis **70** are at least 0.05% greater than the cross-sectional areas defined by the first and second lobe regions **103**, **104** of the pilot SI joint opening **100**, and the cross-sectional areas defined by the sacrum and ilium guide portions **203**, **204** of pilot SI joint opening **200**.

In some embodiments of the invention, the cross-sectional areas of the regions defined by the first and second elongated partially cylindrical sections **76a**, **76b** of the prosthesis **70** are substantially equal to or slightly smaller, e.g., $\leq 0.05\%$, than the cross-sectional areas defined by the first and second lobe regions **103**, **104** of the pilot SI joint opening **100**, and the cross-sectional areas defined by the sacrum and ilium guide portions **203**, **204** of pilot SI joint opening **200**.

According to the invention, the prosthesis **70**, as well as the prostheses disclosed in Co-pending priority application Ser. No. 13/857,977, can comprise various biocompatible materials, including metals and metal alloys, such as titanium, stainless-steel, cobalt-chromium alloys and nickel-titanium alloys.

The prosthesis **70**, as well as the prostheses disclosed in Co-pending priority application Ser. No. 13/857,977, can

also comprise various biocompatible polymers, including, without limitation, reinforced polymers, such as carbon fiber reinforced polymers and metal-framed polymers.

According to the invention, the prosthesis 70, as well as the prostheses disclosed in Co-pending priority application Ser. No. 13/857,977, can also comprise a porous structure to facilitate (i) adhesion of the prosthesis 70 to a post-prosthesis insertion SI joint opening of the invention; particularly, post-prosthesis insertion SI joint openings 300, 400 and, thereby, to SI joint bone structures, i.e., sacrum and ilium bone structures, and (ii) bone or osseous tissue ingrowth into the prosthesis 70.

According to the invention, the prosthesis 70, as well as the prostheses disclosed in Co-pending priority application Ser. No. 13/857,977, can also comprise various exterior surface textures and roughness to facilitate or enhance engagement of the prosthesis to a post-prosthesis insertion SI joint opening, such as post-prosthesis insertion SI joint openings 300, 400, and, thereby, to SI joint bone structures, i.e., sacrum and ilium bone structures, and/or maintain engagement thereto and positioning therein. The surface of the prosthesis 70 (as well as the prostheses disclosed in Co-pending priority application Ser. No. 13/857,977) can, thus, comprise a roughness grade number of N1 (Ra=0.025 μ m), N2 (Ra=0.05 μ m), N3 (Ra=0.1 μ m), N4 (Ra=0.2 μ m), N5 (Ra=0.4 μ m), N6 (Ra=0.08 μ m), N7 (Ra=1.6 μ m), N8 (Ra=3.2 μ m), N9 (Ra=6.3 μ m), N10 (Ra=12.5 μ m), N11 (Ra=25 μ m) or N12 (Ra=50 μ m).

In some embodiments of the invention, the prosthesis 70, as well as the prostheses disclosed in Co-pending priority application Ser. No. 13/857,977, further comprise an outer coating.

In some embodiments, the outer coating comprises a biocompatible and, preferably, biodegradable adhesive composition. According to the invention, suitable adhesive compositions include, without limitation, poly(L-glutamic acid)-based compositions, poly(γ -glutamic acid)-based compositions, poly(alkyl cyano acrylate)-based compositions, polyacrylic acid-based compositions, including polyacrylic acid crosslinked with pentaerythritol and/or allyl sucrose, polyacrylic acid crosslinked with divinyl glycol and combinations thereof; fibrin-based compositions, collagen-based compositions, including collagen and poly(L-glutamic acid) compositions; albumin-based compositions, including BioGlue® (comprises purified bovine serum albumin (BSA) and glutaraldehyde); cyanoacrylate compositions, including butyl-2-cyanoacrylate adhesives (e.g., Indermil®, Histoacryl®, Histoacryl® Blue, and LiquiBand®) and octyl-2-cyanoacrylate adhesives (e.g., Dermabond®, SurgiSeal™, LiquiBand® Flex, and Octyl-Seal); polyethylene glycol (PEG) based compositions, including FocalSeal®, Progel™, Duraseal™, DuraSeal™ Xact, Coseal® and ReSure Sealant; polysaccharide-based compositions, polypeptide-based compositions, and radiation curable materials, such as poly(glycerol-co-sebacate) acrylate (PGSA), discussed below.

In some embodiments, the outer coating comprises a biologically active composition comprising one of the aforementioned biologically active agents (referred to generally as fixation catalysts in Co-pending priority application Ser. No. 13/857,977) or a pharmacological composition comprising one of the aforementioned pharmacological agents.

In some embodiments, the outer coating comprises one of the aforementioned polymers and/or compositions comprising same.

In some embodiments, the aforementioned polymer compositions comprise one or more of the aforementioned biologically active agents or pharmacological agents.

In some embodiments of the invention, the polymer comprises poly(glycerol sebacate) (PGS) or a derivative thereof, including, without limitation, poly(glycerol-co-sebacate) acrylate (PGSA) and PGS co-polymers, such as poly(glycerol sebacate)-co-poly(ethylene glycol) (PGS-PEG); and/or composites thereof, e.g., PGS-hydroxyapatite (HA) composites and PGS-poly(ϵ -caprolactone) (PGS-PCL) composites, and compositions comprising same.

As set forth in Co-pending U.S. application Ser. No. 17/463,779, PGS and derivatives thereof possess a unique property of inducing remodeling of damaged osseous or bone tissue, such as at pilot SI joint openings, and, hence, healing of the associated bone structures when disposed proximate thereto.

As set forth in Loh, et al., *Poly(glycerol sebacate) Biomedical: Synthesis and Biomedical Applications*, Journal of Materials Chemistry B, vol. 3(39), pp. 7641-7652 (2015) and indicated in Table 1 below, a further seminal property of PGS is that its physical state can be modulated during synthesis by controlling the “degree of esterification” via at least one crosslinking agent, e.g., methylene diphenyl diisocyanate (MDI).

TABLE 1

Degree of Esterification	Physical State
$\leq 46\%$	Solid (Brittle Wax)
$\sim 47\%$ -64%	Semi-Solid (Soft Wax)
$\sim 65\%$ -75%	Viscous Liquid
$\sim 76\%$ -83%	Sticky Elastomer
$\geq 84\%$	Elastomer

According to the invention, any suitable degree of esterification of PGS can be employed for PGS when employed in or for PGS based outer coatings (i.e., polymer compositions comprising PGS) and biologically active agent compositions of the invention.

In some embodiments, the PGS based outer coatings comprise a degree of esterification in the range of $\sim 76\%$ -83%, whereby the PGS exhibits adhesive properties, which will enhance engagement of prosthesis 70 (as well as the prostheses disclosed in Co-pending priority application Ser. No. 13/857,977) to the post-prosthesis insertion SI joint openings 300, 400 and, thereby, to the SI joint bone structures, i.e., sacrum and ilium bone structures.

As is well established, the physical state of poly(glycerol-co-sebacate) acrylate (PGSA) can also be modulated by combining the PGSA with a suitable photoinitiator and subjecting the PGSA to radiation.

Indeed, as set forth in Nijst, et al., *Synthesis and Characterization of Photocurable Elastomers from Poly(Glycerol-Co-Sebacate)*, Biomacromolecules, vol. 8, no. 10, pp. 3067-3073 (2007), PGSA can be induced to transition from a liquid or flowable state to a solid elastomer state when combined with a photoinitiator, such as 2-hydroxy-1-[4-hydroxyethoxy] phenyl]-2-methyl-1-propanone (D 2959, Ciba Geigy), 2,2-dimethoxy-2-phenylacetophenone, titanocenes, fluorinated diaryltitanocenes, iron arene complexes, manganese decacarbonyl and methylcyclopentadienyl manganese tricarbonyl, and subjected to radiation, such as visible light; particularly, radiation in the range of approximately 380-750 nm, and ultraviolet (UV) light, particularly, radiation in the range of 10-400 nm.

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Thus, in some embodiments, a composition comprising PGSA (also referred to herein as a “PGSA based composition” and “fixation composition”) is employed to enhance the engagement of the prosthesis 70 to a post-prosthesis insertion SI joint opening, such as post-prosthesis insertion SI joint openings 300, 400, and, thereby, SI joint bone structures, i.e., sacrum and ilium bone structures.

In such embodiments, the PGSA based composition (in a flowable state) is disposed in the internal prosthesis engagement member lumens 86a, 86b of the prosthesis 70, whereby the PGSA based composition is dispersed when the prosthesis 70 is positioned in the dysfunctional SI joint and fills any gaps between the prosthesis 70 and a post-prosthesis insertion SI joint opening of the invention; particularly, post-prosthesis insertion SI joint openings 300, 400, and thereafter cured via radiation and solidified, whereby the solidified PGSA enhances the engagement of the prosthesis 70 to the post-prosthesis insertion SI joint opening and, thereby, to the sacrum and ilium bone structures.

PGS and its derivatives; particularly, PGSA are also excellent platforms for delivery and, hence, administration of biologically active agents and pharmacological agents to mammalian tissue, including osseous or bone tissue.

Thus, in some embodiments of the invention, the PGS based outer coatings and PGS and PGSA based compositions further comprise one or more of the aforementioned biologically active or pharmacological agents.

As indicated above, in some embodiments of the invention, the system for stabilizing dysfunctional SI joints further comprises an image capture apparatus configured and adapted to capture images reflecting positions and/or orientations of the elongated guide probe and/or defect creation assembly when disposed in the body, and, particularly, during advancement of the elongated guide probe and defect creation assembly toward and into the dysfunctional SI joint.

According to the invention, suitable image capture apparatus comprise a fluoroscope, a CT system, an ultrasound system, a radiography system, and a magnetic resonance imaging system.

As also indicated above, in some embodiments of the invention, the system for stabilizing dysfunctional SI joints further comprises a drill guide assembly that facilitates proper placement of (i) the elongated guide probe 20 in the dysfunctional SI joint, and (ii) the pilot SI joint openings 100, 200 of the invention and, hence, sacrum and ilium portions thereof, and, thereby, placement of the prosthesis 70 (and other prostheses described in Co-pending U.S. application Ser. No. 13/857,977) in the dysfunctional SI joint.

Referring now to FIG. 9A, there is shown a preferred embodiment of a drill guide assembly 500 of the invention.

As illustrated in FIG. 9A, the drill guide assembly 500 comprises an access sleeve 502, drill guide 520 and a guide pin 530.

Referring now to FIGS. 9B-9D, there is shown a preferred embodiment of the access sleeve 502.

As illustrated in FIGS. 9B-9D, the access sleeve 502 comprises proximal and distal ends 504a, 504b, and an internal opening 506 that extends from the proximal end 504a to the distal end 504b of the access sleeve 502, and a plurality of lumens 507, which, as illustrated in FIG. 9A, are sized and configured to receive and position Kirschner wires (K-wires) 509 or similar pin structures therein.

As illustrated in FIG. 9A, in a preferred embodiment, the access sleeve internal opening 506 is sized and configured to receive and position the drill guide 520 therein.

As further illustrated in FIGS. 9B and 9D, the proximal end 504a of the access sleeve 502 comprises a planar region

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503, which, as illustrated in FIG. 9A, is configured to seat the proximal end 522a of the drill guide 520 (discussed below) thereon.

In a preferred embodiment, as additionally shown in FIGS. 9B and 9D, the proximal end 504a of the access sleeve 502, i.e., planar region 503, further comprises two (2) threaded holes 505a, 505b, which are preferably disposed on opposing edge regions of the planar region 503. According to the invention, the threaded holes 505a, 505b are sized and configured to receive the threaded end 514 of the access sleeve handle 510, discussed below.

Referring now to FIGS. 9E and 9F, there is shown a preferred embodiment of the access sleeve handle 510.

As illustrated in FIGS. 9E and 9F, the access sleeve handle 510 preferably comprises an elongated cylindrical shaped member comprising proximal and distal ends 512a, 512b.

As further illustrated in FIG. 9E, in a preferred embodiment, the distal end 512b of the access sleeve handle 510 comprises a threaded extension 514 that is sized and configured to cooperate with the threaded holes 505a, 505b of the access sleeve 502, whereby the access sleeve handle 510 can be threadably engaged to the access sleeve 502.

Referring now to FIGS. 9G-9J, there is shown a preferred embodiment of the drill guide 520.

As illustrated in FIGS. 9E-9J, the drill guide 520 comprises proximal and distal ends 522a, 522b, a pair of drill guide lumens 524a, 524b and a drill guide medial lumen 526; the drill guide lumens 524a, 524b and drill guide medial lumen 526 extending from the proximal end 522a to the distal end 522b of the drill guide 520.

As illustrated in FIG. 9A, in a preferred embodiment, the drill guide lumens 524a, 524b are sized and configured to receive (i) a bone dislodging system 40 of the defect creation assembly 30, in this instance, the drill bit 41 shown in FIG. 9L, and (ii) the guide pin 530 shown in FIG. 9K and discussed below.

In a preferred embodiment, the drill guide medial lumen 526 is sized and configured to receive and guide the elongated guide probe 20 of the invention to a desired position proximate the dysfunctional SI joint.

According to the invention, the drill guide internal lumens 524a, 524b and drill guide medial lumen 526 can also be sized and configured to receive various other suitable instruments, such as surgical scopes, center punches, location pins, drill probes and drill stop assemblies, to facilitate the creation of a pilot SI joint opening.

Referring back to FIGS. 9G and 9H, in a preferred embodiment, the proximal end 522a of the drill guide 520 comprises a planar configuration comprising an extended region 523, which, as illustrated in FIG. 9A, is sized and configured to abut the proximal end 504a of the access sleeve 502 to position the drill guide 520 therein.

Referring now to FIG. 9K, there is shown one embodiment of a guide pin 530 of the invention.

As illustrated in FIG. 9K, the guide pin 530 preferably comprises an elongated guide member 532 comprising proximal and distal ends 534a, 534b. The guide pin 530 further comprises a handle 536 that is operatively connected to the proximal end 534a of the guide member 532.

As further indicated above, in some embodiments, the system for stabilizing dysfunctional SI joints further comprises a prosthesis extraction assembly 600 that is configured and adapted to remove the prosthesis 70 from the expanded post-prosthesis insertion SI joint opening and, thereby, dysfunctional SI joint.

Referring now to FIGS. 10A-10G, there is shown a preferred embodiment of a prosthesis extraction assembly 600 of the invention with prosthesis 70 connected thereto.

As illustrated in FIG. 10A, the prosthesis extraction assembly 600 generally comprises prosthesis extraction rods or screws 602a, 602b, an extraction fork 610 and a slap hammer assembly 620.

As illustrated in FIG. 10A-10C, the prosthesis extraction rods 602a, 602b comprise elongated rod members 605 comprising capped proximal ends 604a and threaded distal ends 604b, which are sized and configured to threadably engage threaded internal prosthesis engagement lumens 86a, 86b of the prosthesis 70.

As illustrated in FIGS. 10D-10F, the extraction fork 610 comprises proximal and distal ends 614a, 614b, a primary recess 612 and secondary recesses 615 proximate the distal end 614b. The extraction fork 610 further comprises a threaded lumen 616 proximate the proximal end 614a, which, as discussed below, is sized and configured to threadably engage the threaded distal end 624b of the elongated rod member 621 of the slap hammer assembly 620 discussed below.

As illustrated in FIG. 10A, in a preferred embodiment, the secondary recesses 615 of the extraction fork 610 are configured to receive and releasably engage or ensnare the capped proximal ends 604a of the prosthesis extraction rods 602a, 602b.

As further illustrated in FIG. 10A, the extraction fork 610 is further configured to releasably engage the threaded distal end 624b of the slap hammer assembly 620 via the threaded lumen 616.

As illustrated in FIGS. 10A and 10G, the slap hammer assembly 620 comprises a handle 622, an elongated rod member 621, a weighted sleeve member 626 and a bump stop 628.

As illustrated in FIG. 10G, the elongated rod member 621 comprises a proximal end 624a, comprising internal threads 625, and a threaded distal end 624b.

As further illustrated in FIG. 10G, the handle 622 comprises threaded distal end 623, which, as illustrated in FIG. 10A, is sized and configured to threadably engage the proximal end 624a of the elongated rod member 621.

As illustrated in FIG. 10A, the threaded distal end 623 of the handle 622 is further sized and configured to receive and seat the bump stop 628 thereon, wherein the bump stop 628 is securely positioned between the handle 622 and elongated rod member 621 when the handle 622 is engaged to the elongated rod member 621.

In a preferred embodiment, the weighted sleeve member 626 is configured to slidably translate along the elongated rod member 621 and about the proximal end 614a of the extraction fork 610 and bump stop 628 when the slap hammer assembly 620 is releasably engaged to the extraction fork 610.

According to the invention, removal of prosthesis 70 from the expanded post-prosthesis insertion SI joint opening is achieved as follows:

the prosthesis extraction rods 602a, 602b are initially connected to prosthesis 70; and

after the prosthesis extraction rods 602a, 602b are connected to prosthesis 70, the surgeon grips the handle 622 of the prosthesis extraction assembly 600 and forcibly abuts the handle 622 against the bump stop 628, wherein a removal or extraction force is exerted on prosthesis 70 via the prosthesis extraction rods 602a, 602b and the prosthesis 70 is released from the expanded post-prosthesis insertion SI joint opening.

The following example is provided to enable those skilled in the art to more clearly understand and practice the present invention. The example should not be considered as limiting the scope of the invention, but merely as being illustrated as representative thereof.

An adult male, age 42 presented with a traumatic injury proximate the SI joint, resulting in a dysfunctional SI joint and significant pain associated therewith, i.e., a visual analog pain score (VAS) of approximately 8.0.

A CT scan was initially performed to determine the full extent of the patient's injury, check for any SI joint abnormalities and plan the stabilization procedure, including the prosthesis structure required to stabilize the dysfunctional SI joint.

The stabilization procedure was performed in accord with the method set forth in Co-pending U.S. application Ser. No. 17/463,779; specifically, ¶¶ [000258]-[000261] thereof. The specifics of the procedure were as follows:

Prosthesis

The prosthesis selected for the procedure was similar to prosthesis 70 illustrated in FIGS. 6A-6H and described in detail above. The prosthesis comprised a length of 30 mm and the elongated partially cylindrical sections, i.e., barrels, of the prosthesis comprised a diameter of 7.5 mm. The prosthesis was sourced from Applicant, i.e., Tenon Medical, Inc., and referred to as a CATAMARAN SIJ Fixation System™.

The prosthesis included a bone graft material, which was placed in the barrels of the prosthesis after the prosthesis was implanted in the dysfunctional SI joint.

Posterior Inferior Surgical Approach

The initial incision was placed along the lateral lip of the posterior third of the iliac crest to the posterior superior spine, which provided a prosthesis entry point into the dysfunctional SI joint through the posterior ligaments at approximately the S3 level. The trajectory of the prosthesis was toward the mid-point of the S1 end plate and the sacral promontory.

Creation of Pilot SI Joint Opening

The pilot SI joint opening was created with the defect creation assembly shown in FIGS. 3B and 3C, and described above. The bone dislodging apparatus of the assembly comprised a drill assembly and associated drill bit.

The pilot SI joint opening was similar to pilot SI joint opening 200 described above. The pilot SI joint opening was created by drilling a first opening in the sacrum bone structure and a second opening in the ilium bone structure (such as shown in FIG. 4B) with the drill assembly.

Radiological Assessment

CT scan images of the patient's SI joint six (6) months after the SI joint stabilization procedure, which are shown in FIGS. 11A and 11B, reflect (i) secure and proper placement of the prosthesis in the SI joint, (ii) substantial solid bridging of osseous tissue, and, hence, bone across the SI joint and, (iii) substantial ossification around the prosthesis.

Post-Procedure SI Joint Pain Relief and Function

After a recovery period of fourteen (14) days, the patient reported that the pain had been substantially reduced. Indeed, as shown in FIG. 12, fourteen (14) days after the procedure a visual analog pain score (VAS) of 1.8 was achieved.

The patient was also subjected to a series of post procedure tests to determine the stability of the SI joint and mobility of the musculoskeletal structures of the pelvic and lumbar regions proximate the SI joint. The results were very

favorable. The patient tested positive to the flexion abduction and external rotation (FABER) test. The patient also responded very favorably to Gaenslen, thigh thrust, compression and distraction tests.

The tests thus confirmed that the post procedure SI joint was stabilized and that the musculoskeletal structures of the pelvic and lumbar regions proximate thereto were restored to a near normal level.

As will readily be appreciated by one having ordinary skill in the art, the present invention provides numerous advantages compared to prior art methods and apparatus for stabilizing dysfunctional SI joints. Among the advantages are the following:

- the provision of improved SI joint stabilization systems and apparatus, which can be readily employed in minimally-invasive SI joint stabilization procedures to stabilize dysfunctional SI joints via a posterior approach;
- the provision of improved SI joint prostheses, which, when implanted in a dysfunctional SI joint, effectively ameliorate pain associated with the SI joint dysfunction;
- the provision of improved SI joint prostheses, which can readily be employed in minimally-invasive SI joint stabilization procedures and provide secure engagement to SI joint structures;
- the provision of improved SI joint prostheses, which can readily be employed in minimally-invasive SI joint stabilization procedures and possess optimal structural properties to effectively stabilize dysfunctional SI joints; and
- the provision of improved SI joint prostheses, which can readily be employed in minimally-invasive SI joint stabilization methods and facilitate remodeling of damaged osseous tissue and regeneration of new osseous tissue and osseous tissue structures.

Without departing from the spirit and scope of this invention, one of ordinary skill can make various changes and modifications to the invention to adapt it to various usages and conditions. As such, these changes and modifications are properly, equitably, and intended to be, within the full range of equivalence of the following claims.

What is claimed is:

1. A system for stabilizing a dysfunctional sacroiliac (SI) joint of a subject, said dysfunctional SI joint disposed between and defined by a sacrum bone structure and an ilium bone structure, said system comprising:

- an elongated guide probe, a defect creation assembly, a prosthesis and a prosthesis deployment assembly, said defect creation assembly adapted to access said dysfunctional SI joint in a posterior trajectory,
- said elongated guide probe adapted to be advanced into said dysfunctional SI joint,
- said defect creation assembly adapted to create a pilot SI joint opening in said dysfunctional SI joint, said pilot SI joint opening comprising a sacrum structure opening and an ilium structure opening,
- said defect creation assembly comprising a defect creation assembly proximal end, a defect creation assembly distal end disposed opposite said defect creation assembly proximal end, and a guide probe lumen that extends through said defect creation assembly, said guide probe lumen adapted to receive said guide probe therein,
- said defect creation assembly further comprising a bone dislodging member adapted to dislodge portions of bone in said dysfunctional SI joint,
- said prosthesis sized and adapted to be advanced into said pilot SI joint opening in said posterior trajectory, said

prosthesis comprising a hollow monolithic member, said monolithic member comprising a first elongated section, a second elongated section, and a bridge section, said bridge section disposed between said first elongated section and said second elongated section in any direction,

said monolithic member further comprising a first longitudinal axis,

said first elongated section configured to be advanced into said sacrum bone structure when said prosthesis is said advanced into said dysfunctional SI joint in said posterior trajectory,

said second elongated section configured to be advanced into said ilium bone structure when said prosthesis is said advanced into said dysfunctional SI joint in said posterior trajectory,

said bridge section configured to be advanced into said intraarticular region of said dysfunctional SI joint when said prosthesis is said advanced into said dysfunctional SI joint in said posterior trajectory,

said first elongated section comprising a first open proximal end, a first closed distal end and a first length from said first open proximal end to said first closed distal end,

said first elongated section further comprising a first internal elongate lumen extending from said first open proximal end to said first closed distal end, said first internal elongate lumen comprising a second longitudinal axis and first internal threads extending from said first open proximal end of said first elongated section,

said first elongated section further comprising a first plurality of slots, said first plurality of slots in communication with said first elongate lumen of said first elongated section,

said first elongated section further comprising a first tapered region disposed on said first closed distal end, said second elongated section comprising a second open proximal end, a second closed distal end and a second length from said second open proximal end to said second closed distal end,

said second elongated section further comprising a second internal elongate lumen extending from said second open proximal end to said second closed distal end, said second internal elongate lumen comprising a third longitudinal axis and second internal threads extending from said second open proximal end of said second elongated section,

said second elongated section further comprising a second plurality of slots, said second plurality of slots in communication with said second elongate lumen of said second elongated section,

said second elongated section further comprising a second tapered region disposed on said second closed distal end,

said bridge section comprising a fourth longitudinal axis, a closed proximal end, a third closed distal end, and a third length from said closed proximal end to said third closed distal end, said third length of said bridge section being less than said first length of said first elongated section and said second length of said second elongated section,

said bridge section further comprising an elongate opening disposed between said closed proximal end and said third closed distal end of said bridge section and extending through said bridge section, said elongate opening being in communication with said first plurality of slots and, thereby said first elongate lumen of said

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first elongated section and said second plurality of slots and, thereby said second elongate lumen of said second elongated section,

said bridge section further comprising a third tapered region disposed on said third distal end, said third tapered region comprising a gradually decreasing thickness along said fourth longitudinal axis of said bridge section, said fourth longitudinal axis of said bridge section being coincident with said first longitudinal axis of said monolithic member,

said third tapered region comprising a first tapered section and an opposing second tapered section, said first tapered section and said second tapered section intersecting at said third closed distal end of said bridge section,

said prosthesis deployment assembly configured and adapted to engage said prosthesis and advance said prosthesis into said pilot SI joint opening in said posterior trajectory,

said prosthesis deployment assembly comprising an elongated guide member and a prosthesis engagement rod, said elongated guide member comprising a guide member proximal end and a guide member distal end,

said prosthesis deployment assembly further comprising a guide member lumen that extends through said elongated guide member from said guide member proximal end to said guide member distal end, said guide member lumen sized and configured to receive said prosthesis engagement rod therein,

said prosthesis engagement rod comprising an engagement rod proximal end and an engagement rod distal end, said engagement rod distal end comprising first external threads,

said prosthesis engagement rod sized and configured to be first advanced into said first internal elongate lumen of

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said prosthesis and second advanced into said second internal elongate lumen of said prosthesis,

said first external threads of said prosthesis engagement rod sized and configured to cooperate with said first internal threads of said first internal elongate lumen of said prosthesis when said prosthesis engagement rod is said first advanced into said first internal elongate lumen, and cooperate with said second internal threads of said second internal elongate lumen of said prosthesis when said prosthesis engagement rod is said second advanced into said second elongate lumen of said prosthesis,

said elongated guide member further comprising a prosthesis guide pin that extends from said guide member distal end, said prosthesis guide pin sized and configured to be third advanced into said first internal elongate lumen of said prosthesis and fourth advanced into said second internal elongate lumen of said prosthesis.

2. The system of claim 1, wherein said prosthesis further comprises a biologically active composition, said biologically active composition being disposed in at least one of said first internal elongate lumen and said second internal elongate lumen of said prosthesis.

3. The system of claim 2, wherein said biologically active composition comprises a biologically active agent selected from the group consisting of demineralized bone matrix, autograft bone material, allograft bone material, xenograft bone material, polymethyl-methacrylate, calcium-based bone void filler material, including hydroxyapatite (HA) and tricalcium phosphate.

4. The system of claim 2, wherein said biologically active composition comprises a bone morphogenic protein (BMP) selected from the group consisting of BMP-1, BMP2a, BMP2b, BMP3, BMP4, BMP5, BMP6, BMP7 and BMP8a.

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